



The Risks of Climate Change to the United States in the 21st Century



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At a Glance

The economic effects of climate change will depend on the extent of its physical effects. Those effects are highly uncertain. The Congressional Budget Office estimates that by 2100, there is a 5 percent chance that average global temperatures will be more than 4 degrees Celsius (4°C) warmer than they were in the latter half of the 19th century and an equal chance that they will have risen by less than 2°C. In the United States, sea levels have a 5 percent chance of rising by about 4 feet or more by 2100 and an equal chance of rising by about 2 feet or less. Damage from natural disasters is also expected to increase.

The uncertainty of climate change's physical effects implies a wide range of possible economic consequences, ranging from benign to catastrophic. In this report, CBO focuses, where possible, on the 5th and 95th percentiles of the distributions of potential outcomes. The report examines the possible economic effects of climate change on gross domestic product (GDP), real estate markets, and other areas that influence the economy and the federal budget.

- **Effect on GDP.** In CBO's estimation, there is a 5 percent chance that GDP will be at least 17 percent lower in 2100 than it would have been if temperatures remained stable after 2024 and an equal chance that GDP will be at least 7 percent higher. In CBO's central estimate, GDP would be 3 percent lower than if temperatures remained unchanged.
- **Effects on Real Estate Markets.** CBO estimates that with roughly 4 feet of sea-level rise, recurring flooding would cause the loss of residential property currently worth \$930 billion—roughly 2 percent of the total value. A rise of about 2 feet would cause losses of \$250 billion. Those losses would mostly be borne by property owners, mortgage lenders, insurance companies, and the federal government.
- **Other Consequences.** Climate change will raise mortality rates and rates of illness, especially if increases in temperature are large, though adaptations will mitigate those effects. It will also harm ecosystems that provide food, clean air and water, medicines, and other useful products. Some effects of climate change will increase immigration to the United States and raise defense costs. Hotter temperatures and more intense natural disasters will disproportionately affect low-income households, minority communities, and residents of the Southeast.

The effects of climate change will be more severe if components of the climate system reach certain critical thresholds, sometimes called tipping points. Crossing those thresholds would accelerate greenhouse gas emissions, cause major ice sheets to melt completely, or shift ocean circulation and weather patterns. The temperature changes required to cross those thresholds, the breadth and severity of the effects of crossing them, and the time frames over which those effects would unfold are subject to great uncertainty because they are unprecedented in recorded history.



Contents

Summary	1
How Will Climate Change Affect GDP?	1
How Will Climate Change Affect Real Estate and Financial Markets?	2
What Other Consequences Will Climate Change Have for the Economy and the Federal Budget?	2
Chapter 1: Climate Change Through 2100	5
Sources of Climate Change	5
Projected Changes in Climate	6
Uncertainty and Tipping Points	13
Chapter 2: Effects on Economic Output	19
How Climate Change Affects Output	19
Projected Effects on GDP	20
Chapter 3: Effects on Real Estate and Financial Markets	23
Property at Risk	23
Bearers of Financial Risks Associated With Property Damage	25
Chapter 4: Other Consequences of Climate Change	33
Human Health	33
Biodiversity	35
Immigration	37
National Security	39
Distributional Effects	40
Appendix: Research About Climate Risks to the United States	43
List of Figures	57
About This Document	58
Boxes	
2-1. Limitations of GDP as a Measure of Economic Activity	21
3-1. Transition Risks	28

Notes About This Report

Numbers in the text and figures may not add up to totals because of rounding.

One degree Celsius is 1.8 degrees Fahrenheit.

The Congressional Budget Office has updated this report since its original publication. Changes are listed at the end of the report.

Summary

As emissions of greenhouse gases from human activities accumulate in the atmosphere and oceans, climate conditions are changing throughout the world. In the United States, those changes will have consequences for economic activity, real estate and financial markets, and several other domains. The effects of climate change are highly uncertain, ranging from benign to catastrophic. In this report, to account for that broad range of possible effects, the Congressional Budget Office focuses, where possible, on the 5th and 95th percentiles of the distributions of potential outcomes.

To assess risks to the United States, CBO analyzed a range of climate scenarios from the Intergovernmental Panel on Climate Change. Those scenarios reflect the effects of climate-related policies implemented around the world by the end of 2020, as well as the 2022 reconciliation act (sometimes called the Inflation Reduction Act), which included several provisions intended to reduce emissions. Since the latter half of the 1800s, average global temperatures have increased by more than 1 degree Celsius (1°C). CBO's projections based on those climate scenarios suggest that by 2100, there is a 5 percent chance that warming will exceed 4°C and an equal chance that it will be less than 2°C. The central projection is for roughly 3°C of warming. The full range of potential temperature increases would be lower if further actions were taken to reduce emissions.

Warming will cause changes in global and national precipitation patterns and sea levels. By 2100, U.S. sea levels have a 5 percent chance of rising by about 4 feet or more, on average, and an equal chance of rising by about 2 feet or less, though sea-level rise in different parts of the country will vary considerably. Changes in temperatures, precipitation, and sea levels are expected to increase damage from natural disasters such as hurricanes, floods, wildfires, and droughts.

Those physical effects of climate change would be more severe if components of the climate system reached critical thresholds, sometimes called tipping points. Past those thresholds, greenhouse gas emissions from natural sources would accelerate, major ice sheets would melt

completely, and ocean circulation or weather patterns would distinctly shift. Because projected rates of warming are unprecedented in recorded history, the extent of warming at which such changes would take place, the potential interactions among those changes, and the extent of their effects are subject to considerable uncertainty. Although many of those effects would not occur until after 2100, emissions of greenhouse gases during this century will make them more likely.

Regardless of its severity, climate change will have economic consequences worldwide. Those consequences will be worse for some other countries than for the United States. They will tend to be especially harmful to countries that are less developed or hotter. This report focuses on the effects of climate change on the United States.

How Will Climate Change Affect GDP?

Climate change will have many negative effects on economic activity in the United States, though some of its effects will be positive. Changes in temperature and rainfall will, on average, reduce the productivity and supply of workers—particularly those who work outdoors. Reductions in agricultural productivity in warmer areas will be partially offset by increases in colder ones. Flooding from rising sea levels and more intense hurricanes will exacerbate damage to private businesses, public infrastructure, and people's homes. Efforts to adapt to climate change will moderate those effects, reducing the impact on gross domestic product (GDP).

In CBO's estimation, there is a 5 percent chance that real (inflation-adjusted) GDP in the United States will be at least 17 percent lower in 2100 than it would have been in the absence of additional increases in temperature. There is an equal chance that real GDP in 2100 will be at least 7 percent higher than it would have been otherwise. In 2023, effects of that size would have amounted to a \$4.7 trillion loss and a \$1.9 trillion gain in GDP. Global warming is more likely to decrease GDP than to increase it: The central estimate from CBO's projections is that climate change will reduce real GDP by 3 percent. Even larger losses would be possible if warming pushed components of the climate system past thresholds that

have not been reached in recorded history, but the likelihood of those developments is highly uncertain.

How Will Climate Change Affect Real Estate and Financial Markets?

Throughout the 21st century, climate change will increase the risk of property damage from floods and wildfires. Continued development in vulnerable areas, particularly along the coasts, will compound those risks.

Today, about 17 million properties face at least a 1 percent chance of flooding each year. CBO projects that if U.S. sea levels rose by approximately 4 feet, about \$930 billion in today's residential property would be flooded routinely—roughly twice per month. If U.S. sea levels rose by roughly 2 feet, that damage would total about \$250 billion. Although those amounts are small in relation to the total value of residential properties—\$930 billion corresponds to about 2 percent of property values—the losses will be concentrated along the coasts, particularly affecting economies, households, and mortgage lenders in those areas. Continued real estate development in low-lying coastal areas would increase the amount of property damage due to sea-level rise; more adaptation would reduce it.

Increases in wildfires also pose financial risks. Over the past decade, roughly 2 million properties have faced at least a 1 percent chance of damage from wildfires each year, and the annual costs to homeowners, businesses, and governments have averaged \$10 billion (in 2024 dollars). Hotter and drier conditions are expected to increase the severity of wildfires. According to one study, with 4°C of global warming, wildfire damage will be 10 times greater, and with 3°C of warming, it will be 5 times greater. Adaptations such as implementing better forest management practices and clearing vegetation around homes will mitigate some of that additional damage.

Property owners, mortgage lenders, insurance companies, and the federal government will bear losses due to floods and wildfires. If those losses led to cascading failures of financial institutions, they would undermine the financial system. The likelihood of those developments would depend on the amount of losses, how quickly they were realized, and how concentrated they were in certain areas or among certain lenders. CBO has no basis for assessing the likelihood of that risk.

What Other Consequences Will Climate Change Have for the Economy and the Federal Budget?

Climate change will produce a wide range of other effects with implications for the economy and the federal budget.

Human Health

The effects of climate change on human health and mortality will depend on both the extent of climate change and the extent of adaptations to it. Changes in health and mortality will affect the federal budget, including costs for health care programs and Social Security.

Warmer temperatures are expected to increase deaths on hot days and reduce deaths on cold days. Without further adaptation, warming of 4°C would cause mortality rates in the United States to be 1.5 percent to 2.0 percent higher in 2100 than they would be in the absence of additional increases in temperature. Older people are particularly vulnerable to extreme temperatures, so the more the average age of the population continues to increase, the larger the effect will be. Adaptation, by contrast, will reduce the rise in deaths. Accounting for the effects of adaptation, given the same amount of warming, death rates could either rise or fall by as much as 0.5 percent. Warming of 2°C would have little effect on mortality because the reduction in deaths due to cold weather would roughly offset the increase in deaths due to hot weather.

Climate change will also affect health and mortality through channels other than temperature. For instance, increased pollution from wildfires and expanded ranges of disease-carrying organisms will raise rates of illness and disease.

Biodiversity

Climate change threatens to accelerate losses of biological diversity and alter the ecosystems people depend on for food, clean air and water, and many products. For example, global warming of 4°C would cause over 30 percent of the land currently used for agriculture to become climatically unsuitable for growing crops or raising livestock, though other areas would become more suitable for those purposes. Reductions in global crop yields would make some types of foods scarcer and would increase food prices. With only 2°C of warming, the United States will lose the flood protection provided by its coral reefs, which will increase the costs of flood damage and disaster relief. And because many prescription drugs are developed using molecules from

plants and other organisms, disruption to their ecosystems will affect future medicines and U.S. health care costs. Although animals and plants have some ability to adapt to changing environmental conditions, the pace of warming will alter environments more quickly than many organisms are able to adapt or evolve in response.

Immigration

CBO expects that climate change will make some people more likely to immigrate to the United States, but the extent of any relative rise in immigration is highly uncertain. Agricultural losses caused by droughts and increases in natural disasters are likely to increase existing immigration flows to the United States from adversely affected regions. However, levels of immigration will also depend on administrative and judicial actions and the resources of potential immigrants.

National Security

Climate change is expected to affect national defense operations by making conflict more likely, forcing the military to change elements of its defense strategy, and raising the costs of ongoing operations. Climate change will increase the risk of conflict, including civil war and other unrest within countries and instability among

countries. It will probably also alter the nature of that conflict. For example, as melting ice opens the Arctic, competition for natural resources and shipping channels in the region is likely to intensify, increasing demands on the U.S. military. Finally, greater maintenance requirements at defense installations, damage from natural disasters, and changing needs for heating and cooling on bases will increase the military's operating costs.

Distributional Effects

Communities and regions across the United States will experience the effects of climate change in different ways. Low-income and minority communities will be disproportionately exposed to the physical effects of climate change and will find it hardest to adapt to extreme heat or recover from natural disasters. Residents of the Southeast, the region with the nation's lowest per capita income, will probably be affected by climate change more than residents of other regions. As a result, the budgetary costs for programs that provide cash payments or other assistance to people with low income will probably increase more than they would if the impacts of climate change were distributed evenly.

Chapter 1: Climate Change Through 2100

Human activities—especially fossil fuel use, deforestation, and agriculture—are producing increasing emissions of greenhouse gases and other gases and particulates that affect the climate. Because a large portion of the emitted gases will remain in the atmosphere for centuries, their effects will persist even if emissions decline. According to scientific consensus, the accumulation of greenhouse gases, coupled with widespread changes in land use, is altering climate conditions throughout the world, including in the United States. Since the latter half of the 1800s, average global temperatures have increased by 1.3 degrees Celsius (about 2.3 degrees Fahrenheit).¹

To understand the risks that further climate change would pose to the economy and society, the Congressional Budget Office projected changes in certain key features of the climate through the end of the century. To do so, CBO analyzed projections of future emissions, accounting for the effects of climate policies adopted around the world through 2020, as well as the 2022 reconciliation act. CBO's analysis suggests a wide range of possible outcomes. In particular, there is about a 5 percent chance that by 2100, global warming will exceed 4°C—that is, nearly 3°C more than has already occurred since the 19th century—and a roughly equal chance that warming will be less than 2°C. The central estimate points to about 3°C of warming. Efforts to reduce emissions would lower the range of projected outcomes.

Warming will also lead to changes in precipitation, sea levels, and natural disasters. Many of those changes are reasonably well understood, and their effects can be estimated with confidence. However, because projected levels of warming are unprecedented in recorded history, other possible changes—including changes that could have catastrophic consequences—are poorly understood. The critical thresholds at which such changes would occur, often referred to as tipping points, and the

extent of their effects can be projected only with great uncertainty.

Sources of Climate Change

As energy from the sun is absorbed by the Earth and radiated back into space, greenhouse gases increase the energy held in the lower atmosphere, keeping the Earth's surface warmer than it would otherwise be. Those gases include carbon dioxide (CO₂), methane, nitrous oxide, and several human-made gases containing fluorine and chlorine. Other substances in the atmosphere also affect the climate: Soot absorbs incoming energy and tends to warm the lower atmosphere, whereas other aerosols (such as sulfur dioxide from the combustion of coal) tend to exert offsetting cooling effects.² The internal variability of the climate system and other natural factors, such as the amount of sunlight, have had only minor effects on recent warming.

Historical Emissions

Since the onset of industrialization, human activities have yielded substantial emissions of greenhouse gases, including about 2,500 billion metric tons (gigatons, or Gt) of CO₂. Over half of those CO₂ emissions have been absorbed by land, forests, and oceans, which act as *carbon sinks*, or natural systems that absorb more CO₂ than they release.³ The remaining emissions have raised atmospheric concentrations of CO₂ by more than

1. Copernicus Programme, "Climate Indicators: Temperature" (updated April 22, 2024), <https://tinyurl.com/2mkt83ex>. Estimates of changes in temperature are based on averages across several years to better control for short-term fluctuations.

2. Aerosols from volcanic eruptions and other natural and human-made sources can temporarily result in regional or even global cooling. Those fluctuations are not inconsistent with an ongoing long-term warming trend caused by human activities.

3. On land, vegetation absorbs CO₂ through photosynthesis. In the ocean, physical and chemical processes dissolve CO₂ at the surface of the water, where further chemical reactions convert the dissolved CO₂ into forms of carbon that can remain in the ocean for longer periods. Other greenhouse gases are also absorbed by the land and the ocean but to a much lesser extent. See Josep G. Canadell and others, "Global Carbon and Other Biogeochemical Cycles and Feedbacks," in Valérie Masson-Delmotte and others, eds., *Climate Change 2021: The Physical Science Basis, Working Group I Contribution to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* (Cambridge University Press, 2021), pp. 673–816, <https://doi.org/10.1017/9781009157896>.

50 percent, to levels not seen for about 3 million years. CO₂ emissions account for about 80 percent of the warming caused by greenhouse gases. For other gases, the patterns of accumulation differ but are similarly complex.

Most CO₂ emissions have come from fossil fuel consumption and cement production; a smaller portion have resulted from changes in land use, such as deforestation and the clearing of land for agriculture. Roughly half of the emissions from fossil fuels and cement production have occurred in the past 30 years. Over the past decade, global emissions of CO₂ have averaged about 40 Gt per year.⁴ The United States has produced 13 percent of those annual emissions. About 26 percent have come from China, the world's largest emitter, and about 6 percent have come from India, which ranks third.

Projections of Future Emissions

To analyze the effects of future climate change, CBO drew on the Intergovernmental Panel on Climate Change's sixth assessment report (sometimes called the AR6), a collection of three volumes published between 2021 and 2022 that evaluate and summarize scientific research related to climate change for the United Nations.⁵ The report presents the results of thousands of scenarios of future emissions and climate change based on roughly 200 peer-reviewed modeling frameworks.⁶ CBO analyzed 21 scenarios that incorporated the assumption that climate-related policies implemented around the world by the end of 2020 would be maintained indefinitely and no further policies would be

adopted.⁷ CBO also adjusted the projections to account for the effects of the 2022 reconciliation act, which provided incentives for the development and use of technologies with lower emissions.⁸ (That treatment is consistent with the agency's approach in producing its budgetary and economic projections, which reflect the assumption that current U.S. laws will remain unchanged.)

The scenarios based on implemented policies yield a range of projections of CO₂ emissions over the rest of the century (see Figure 1-1). That range reflects uncertainty about societal factors (such as population growth) as well as physical factors (such as the capacity of sinks to absorb carbon).

In all of the projections involving implemented policies, concentrations of CO₂ in the atmosphere continue to rise through the end of the century, even if emissions start to decline in the coming decades (see Figure 1-2).

Projections of emissions are also subject to uncertainty about future developments that emissions scenarios do not capture. Those developments include policy changes, technological advances, and events such as natural disasters, pandemics, financial crises, or wars. Any of those developments would cause emissions and levels of warming to be higher or lower than projected.

Projected Changes in Climate

The atmospheric accumulation of greenhouse gases and other emissions has initiated an irregular and gradual warming of the Earth's surface and various related changes in precipitation, sea levels, and other aspects of the climate. Tropical and temperate climate zones are expanding, and in North America, growing seasons are lengthening by about one to two days per decade.⁹

4. A metric ton of CO₂ is roughly the amount that an average American household produces in a month by using electricity, heating a home, and driving one vehicle. Measured in terms of the amount of CO₂ that would cause an equivalent amount of warming (CO₂eq), annual emissions of all greenhouse gases combined have averaged about 52 Gt globally and 6 Gt in the United States since 2014. See Hannah Ritchie, Pablo Rosado, and Max Roser, "CO₂ and Greenhouse Gas Emissions" (Our World in Data, 2023), <https://tinyurl.com/3e9p42bn>; and Robbie M. Andrew and Glen P. Peters, "The Global Carbon Project's Fossil CO₂ Emissions Dataset," version 2024v17 (October 23, 2024), <https://doi.org/10.5281/zenodo.13981696>.

5. See Intergovernmental Panel on Climate Change, "Sixth Assessment Report" (accessed October 7, 2024), www.ipcc.ch/assessment-report/ar6.

6. The results of those scenarios are available at International Institute for Applied Systems Analysis, "AR6 Scenario Explorer and Database Hosted by IIASA" (2022), <https://data.ece.iiasa.ac.at/ar6>.

7. Although many countries have signaled an intention or even made a commitment to achieve net zero greenhouse gas or CO₂ emissions by midcentury, few have enacted policies to do so. The set of climate scenarios that CBO analyzed does not incorporate those intentions or commitments.

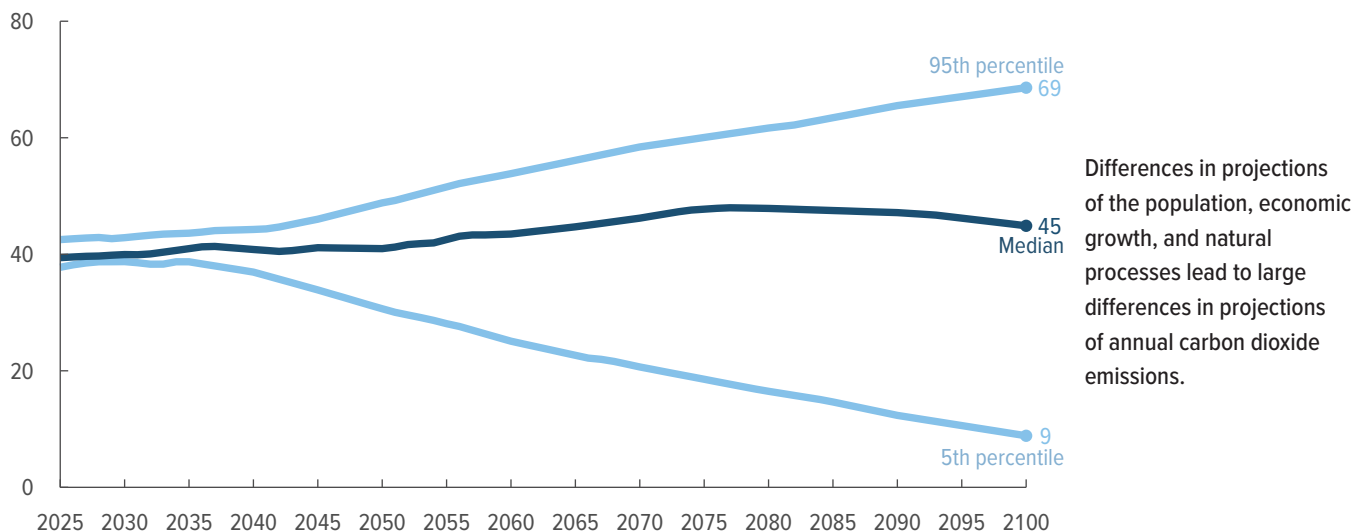
8. CBO used the Environmental Protection Agency's projections of emission reductions under the 2022 reconciliation act to make those adjustments. For more information, see Chad Shirley and William Swanson, *The Effects of Climate Change on GDP in the 21st Century* (Congressional Budget Office, forthcoming).

9. Environmental Protection Agency, "Climate Change Indicators: Length of Growing Season" (updated June 27, 2024), <https://tinyurl.com/4ffcycap>.

Figure 1-1.

Projected Global Carbon Dioxide Emissions Through 2100

Gigatons of carbon dioxide emissions per year



Data source: Congressional Budget Office, using data from the Environmental Protection Agency and the Intergovernmental Panel on Climate Change. See www.cbo.gov/publication/60845#data.

A gigaton is a billion metric tons.

These projections are based on 21 scenarios of future emissions and climate change vetted by the Intergovernmental Panel on Climate Change. The projections incorporate the effects of climate-related policies implemented around the world by the end of 2020 as well as the effects of the 2022 reconciliation act.

Emissions of other greenhouse gases, aerosols, and particulates will also affect climate change. This figure is limited to carbon dioxide, which is by far the most common greenhouse gas, because those other gases and particulates are removed from the atmosphere at very different rates.

Temperature

The average temperature across the globe is currently about 15°C (59°F)—1.3°C (2.3°F) higher than it is estimated to have been in the latter half of the 19th century.¹⁰ Roughly half of that increase has occurred since the mid-1980s.

Rising concentrations of greenhouse gases and changes in land use have caused temperatures to rise. Some of the warming effect from greenhouse gases—most likely about one-quarter—has been offset by the cooling effects of aerosols in the atmosphere. The climate scenarios

described above yield a range of projections of further increases in temperatures over the rest of the century.¹¹ Accounting for the effects of climate policies adopted internationally through 2020 and, in the United States, the 2022 reconciliation act, CBO's central estimate is that by 2100, the average global temperature is likely to be about 3°C warmer than it was during the 1850–1900 period (see Figure 1-3). That difference would amount to an increase of 1.6°C relative to today.

Many uncertain factors could cause increases in temperatures to be larger or smaller than that estimate, including physical processes (such as those associated with emissions, sinks, and aerosols) and future societal changes (such as those involving population levels, economic

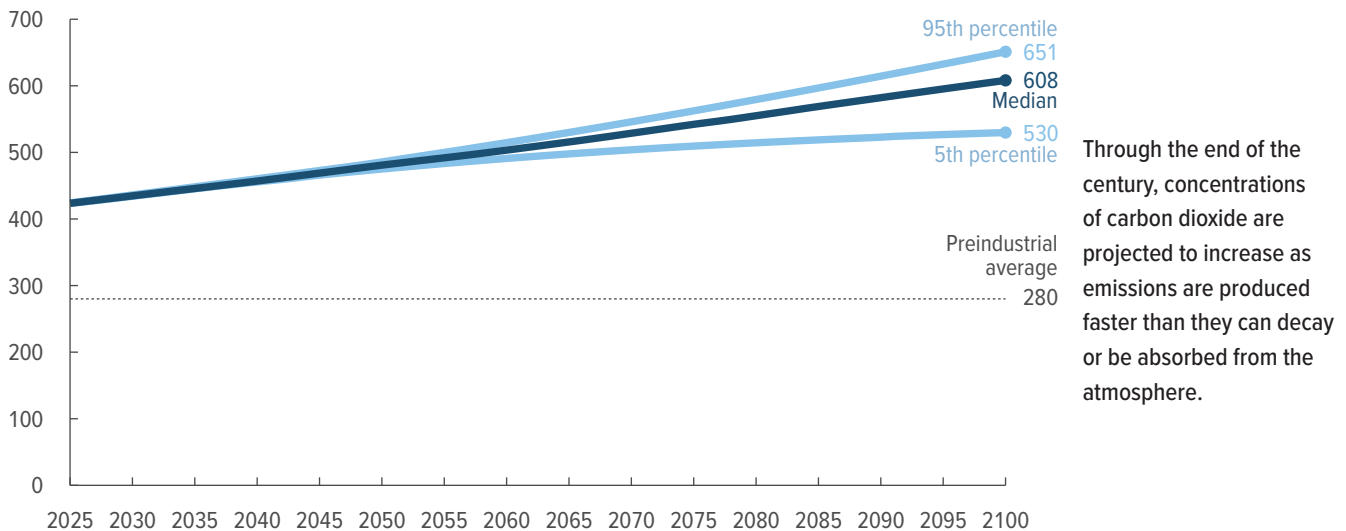
10. The effects of emissions on the climate are thought to have been minimal before 1850. For example, some researchers have estimated that between 1750 and the latter half of the 1800s, average global temperatures increased by 0.1°C. See Deliang Chen and others, "Framing, Context, and Methods," in Valérie Masson-Delmotte and others, eds., *Climate Change 2021: The Physical Science Basis, Working Group I Contribution to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* (Cambridge University Press, 2021), pp. 147–286, <https://doi.org/10.1017/9781009157896>.

11. CBO's temperature projections are based on a weighted average of probability distributions that are fitted to the temperature projections for each of the 21 climate scenarios. For more information, see Chad Shirley and William Swanson, *The Effects of Climate Change on GDP in the 21st Century* (Congressional Budget Office, forthcoming).

Figure 1-2.

Projected Concentrations of Carbon Dioxide in the Atmosphere Through 2100

Carbon dioxide (in parts per million)



Data source: Congressional Budget Office, using data from the Environmental Protection Agency; Integrated Carbon Observation System, “Data Supplement to the Global Carbon Budget 2024,” version 1.0 (2024), <https://doi.org/10.18160/gcp-2024>; the Intergovernmental Panel on Climate Change; and the National Oceanic and Atmospheric Administration. See www.cbo.gov/publication/60845#data.

These projections are based on 21 scenarios of future emissions and climate change vetted by the Intergovernmental Panel on Climate Change. The projections incorporate the effects of climate-related policies implemented around the world by the end of 2020 as well as the effects of the 2022 reconciliation act.

Emissions of other greenhouse gases, aerosols, and particulates will also affect climate change. This figure is limited to carbon dioxide, which is by far the most common greenhouse gas, because those other gases and particulates are removed from the atmosphere at very different rates.

growth, and land use).¹² In CBO’s assessment, there is about a 5 percent chance that the amount of emissions will be large enough and the impact on the climate strong enough that warming exceeds 4°C by 2100. Conversely, there is also about a 5 percent chance that the amount of emissions will be small enough and the impact on the climate moderate enough that warming is less than 2°C by 2100.

In the contiguous United States, the average temperature in 2023 was about 12°C (54°F), but that average masks a wide range across different places, in different seasons, and at different times of day. For any given increase in the average global temperature, temperature increases within the United States will tend to be greater in cooler times and places—greater at night than during the

day, greater in the winter than during the summer, and greater in the northern portions of the country than the southern ones. Inland areas, particularly the West and the South, will experience more warming than coastal areas because oceans absorb and store heat, which moderates temperatures on nearby land (see Figure 1-4).

Precipitation

As a result of climate change, average rainfall over land is increasing at an accelerating rate. In most regions, episodes of heavy rain are becoming more frequent and intense. Within the United States, precipitation is projected to increase in most parts of the country (see Figure 1-5 on page 11). However, in the Southwest and some parts of Texas, precipitation is projected to decline modestly.

Sea Levels

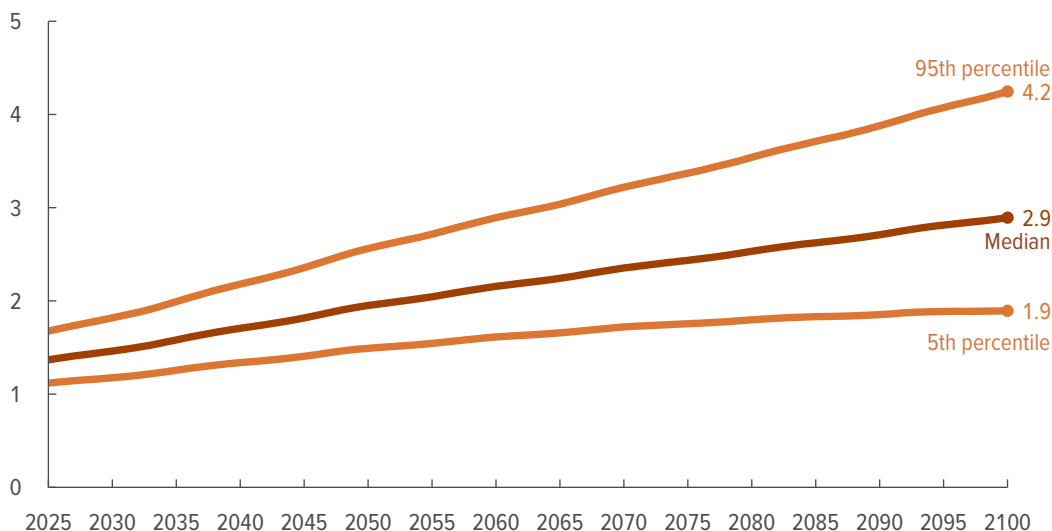
The ocean surface has warmed nearly as much as the atmosphere. As deeper waters gradually warm and expand and as glaciers and ice sheets melt, sea levels are rising at an accelerating pace. Because those processes respond relatively slowly to a warming atmosphere, projected increases in sea levels are fairly consistent

12. Most of the uncertainty surrounding projections of warming over the next 30 years concerns factors in the first category; most of the uncertainty surrounding later projections concerns factors in the second category. See Flavio Lehner and others, “Partitioning Climate Projection Uncertainty With Multiple Large Ensembles and CMIP5/6,” *Earth System Dynamics*, vol. 11, no. 2 (May 2020), pp. 491–508, <https://doi.org/10.5194/esd-11-491-2020>.

Figure 1-3.

Projected Global Warming Relative to the 1850–1900 Period Through 2100

Change in degrees Celsius



By 2100, average global temperatures could be 2°C, 3°C, or 4°C warmer than they were from 1850 to 1900. There is about a 5 percent chance that warming will be less than 2°C and a similar chance that it will exceed 4°C.

Data source: Congressional Budget Office, using data from the Environmental Protection Agency; Integrated Carbon Observation System, “Data Supplement to the Global Carbon Budget 2024,” version 1.0 (2024), <https://doi.org/10.18160/gcp-2024>; the Intergovernmental Panel on Climate Change; and the National Oceanic and Atmospheric Administration. See www.cbo.gov/publication/60845#data.

These projections are based on 21 scenarios of future emissions and climate change vetted by the Intergovernmental Panel on Climate Change. The projections incorporate the effects of climate-related policies implemented around the world by the end of 2020 as well as the effects of the 2022 reconciliation act.

Global average temperatures are based on global surface air temperatures.

across scenarios until later in the century. Even if global temperatures stabilize by 2100, expanding oceans and melting ice will continue to raise sea levels for centuries.

The extent to which sea levels will ultimately rise in response to any amount of warming is highly uncertain, however. Some of that uncertainty reflects a range of potential outcomes due to processes that are well understood, such as the gradual expansion of warming oceans and the melting of glaciers. For example, by 2100, given the central estimate of about 3°C of warming, there is a 5 percent chance that well-understood processes will raise average global sea levels about 3 feet or more above their 2005 levels and an equal chance that the rise will be about 1 foot or less.¹³ The central estimate is an increase of 2 feet. About one more degree of warming would increase those amounts by about 4 inches; one less degree of warming would lower them by a similar amount.

13. William V. Sweet and others, *Global and Regional Sea Level Rise Scenarios for the United States: Updated Mean Projections and Extreme Water Level Probabilities Along U.S. Coastlines*, Technical Report NOS 01 (National Oceanic and Atmospheric Administration, February 2022), <https://tinyurl.com/cmsv48xr>.

Even higher sea levels could result from processes that are poorly understood. Those processes include how warming ocean waters will affect the melting of the bases of ice sheets that lie below sea level. With sufficient warming, such processes could cause global sea-level rise to reach 5 feet or more by 2100—but the likelihood that they will produce that much melting is currently unknown.

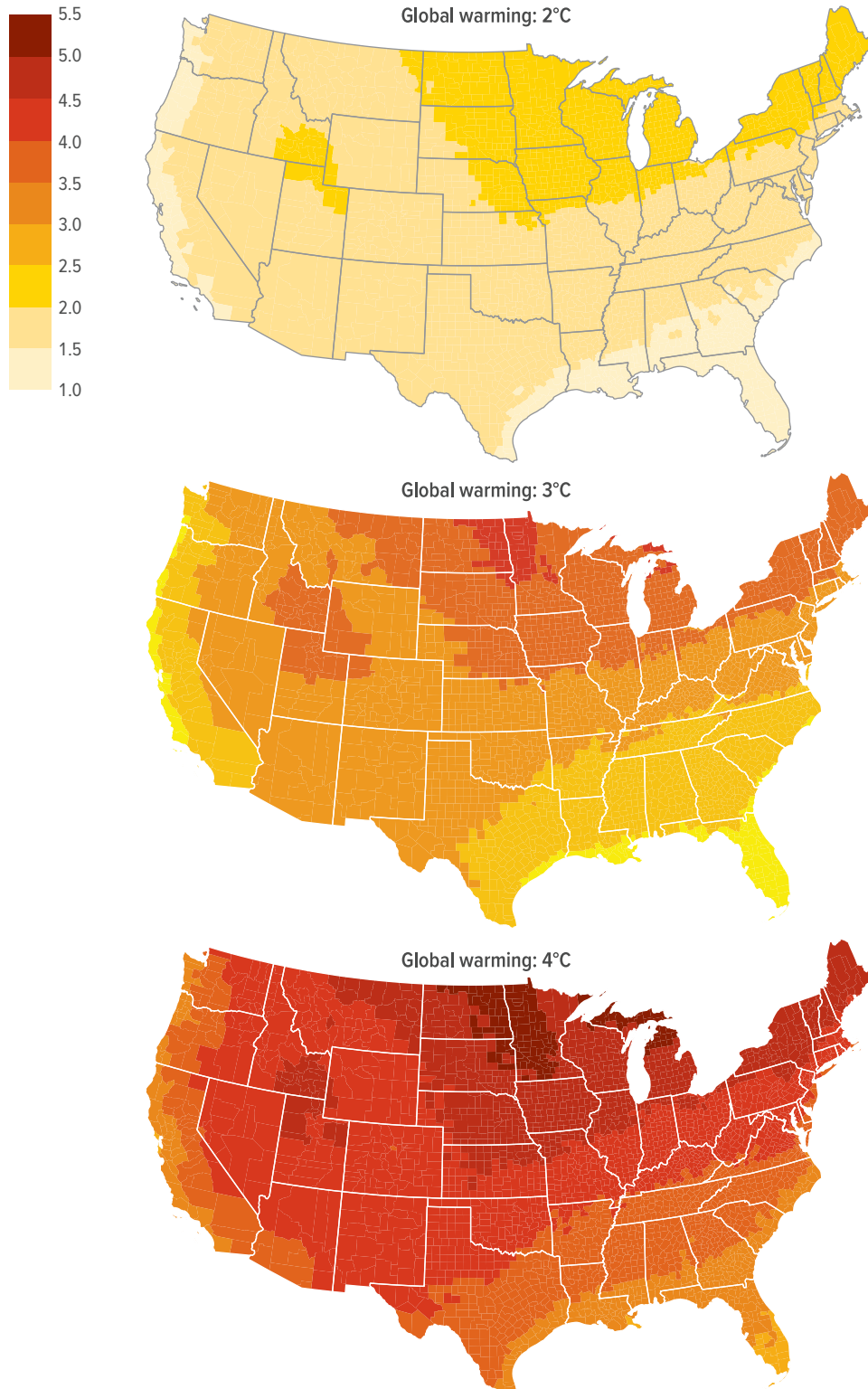
By 2100, average sea levels are projected to rise roughly 8 inches more on the coasts of the contiguous United States than around the globe. With warming of 3°C, that means a 5 percent chance that average U.S. sea levels will rise by about 4 feet or more, another 5 percent chance that they will rise by about 2 feet or less, and a central estimate of approximately 3 feet of sea-level rise. But changes in sea levels will vary substantially across U.S. coastlines (see Figure 1-6 on page 12).¹⁴ Sea levels are expected to rise most along the western Gulf Coast—by roughly 3.5 feet to 5.0 feet, depending on the exact location, given an average increase of about 3 feet nationwide.

14. The factors that cause sea levels to vary from place to place include a host of geological and hydrological processes, such as differences in the subsidence and uplift of land and local ocean circulation patterns.

Figure 1-4.

Projected Increases in Temperatures in the Contiguous United States by 2100, by Amount of Global Warming

Change in degrees Celsius



Northern portions of the country will see the largest temperature increases for any given amount of global warming. Inland areas will warm more than coastal areas.

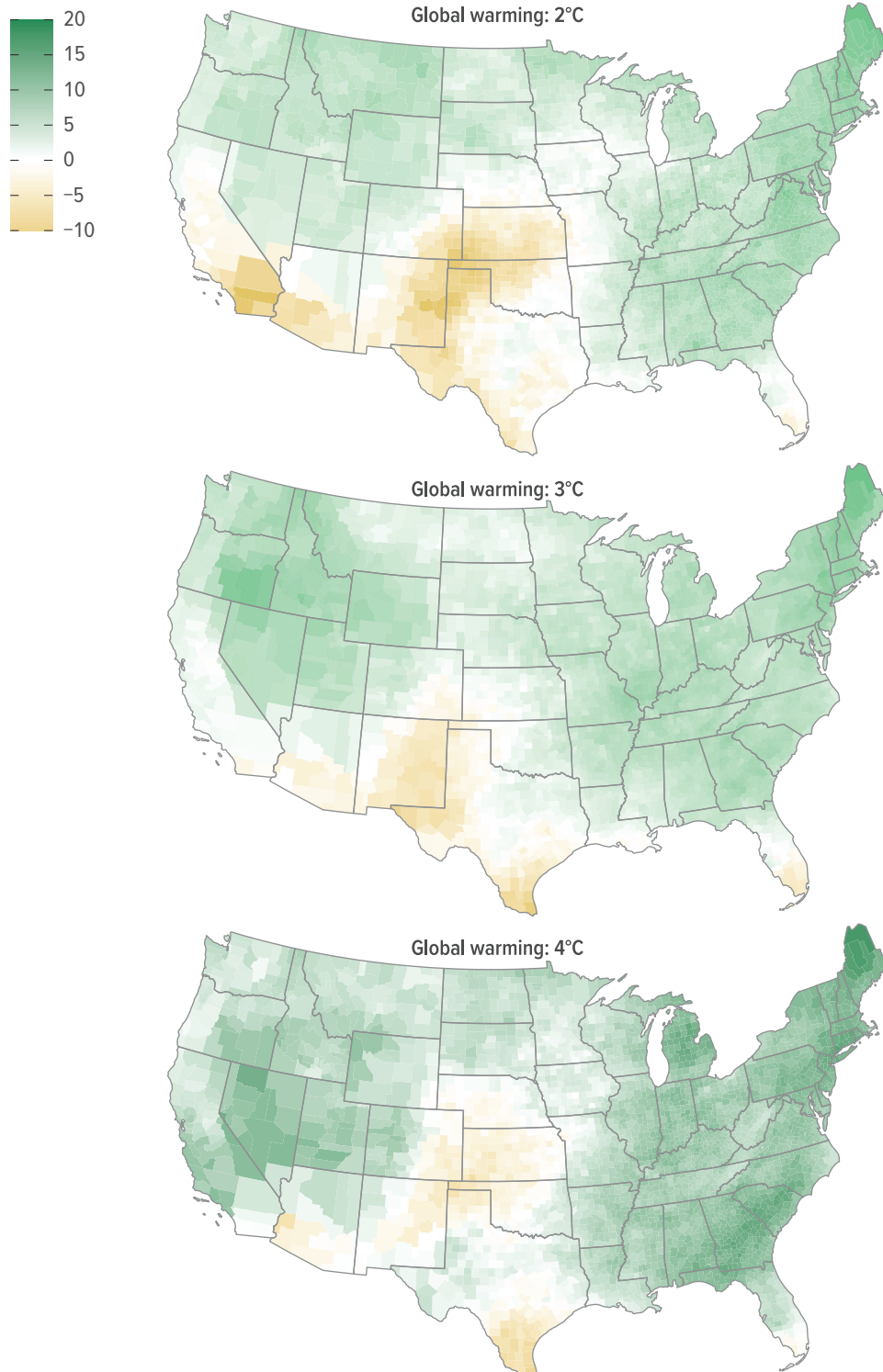
Data source: Congressional Budget Office, using data from Kate Marvel and others, "Climate Trends," in Allison R. Crimmins and others, eds., *Fifth National Climate Assessment* (U.S. Global Change Research Program, November 2023), Chapter 2, <https://nca2023.globalchange.gov/chapter/2>. See www.cbo.gov/publication/60845#data.

By 2100, there is about a 5 percent chance that global warming since the 1850–1900 period will be less than 2°C and a similar chance that it will exceed 4°C.

Figure 1-5.

Projected Changes in Precipitation in the Contiguous United States by 2100, by Amount of Global Warming

Percentage change



Global warming will increase precipitation in most parts of the country, although the Southwest and portions of Texas are projected to experience declines.

Data source: Congressional Budget Office, using data from Kate Marvel and others, "Climate Trends," in Allison R. Crimmins and others, eds., *Fifth National Climate Assessment* (U.S. Global Change Research Program, November 2023), Chapter 2, <https://nca2023.globalchange.gov/chapter/2>. See www.cbo.gov/publication/60845#data.

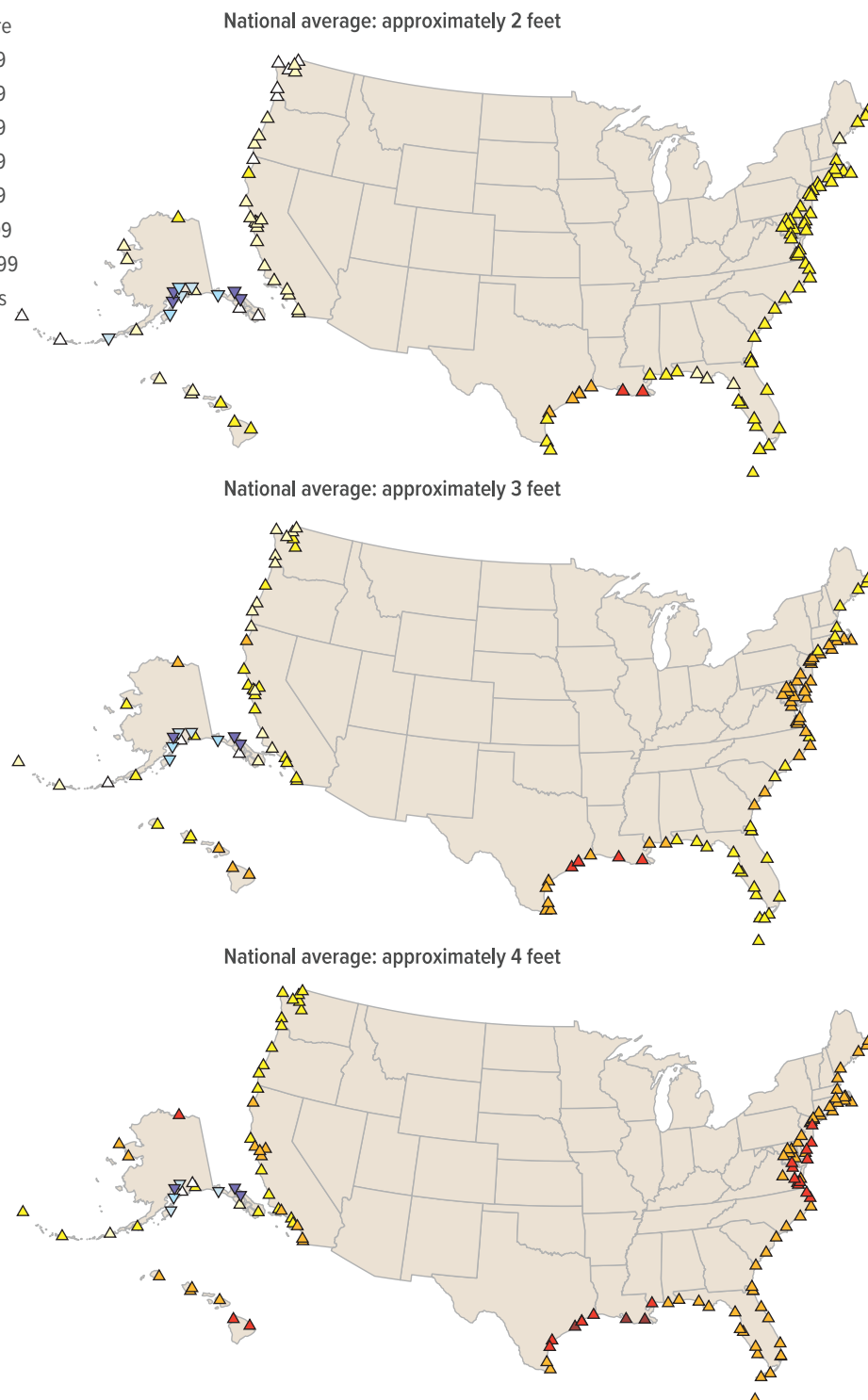
By 2100, there is about a 5 percent chance that global warming since the 1850–1900 period will be less than 2°C and a similar chance that it will exceed 4°C.

Figure 1-6.

Projected Increases in Regional Sea Levels by 2100, by Average U.S. Sea-Level Increase

Change in feet

- ▲ 5 or more
- ▲ 4 to 4.99
- ▲ 3 to 3.99
- ▲ 2 to 2.99
- ▲ 1 to 1.99
- △ 0 to 0.99
- △ 0 to -0.99
- △ -1 to -1.99
- △ -2 or less



Sea-level changes will vary substantially across U.S. coastlines. The largest increases are projected for the western Gulf Coast. Only certain parts of the Alaskan coast are projected to see decreases.

Data source: Congressional Budget Office, using data from Christine L. May and others, "Coastal Effects," in Allison R. Crimmins and others, eds., *Fifth National Climate Assessment* (U.S. Global Change Research Program, November 2023), Chapter 9, <https://nca2023.globalchange.gov/chapter/9>. See www.cbo.gov/publication/60845#data.

The 2-foot, 3-foot, and 4-foot cases roughly correspond to the 5th, 50th, and 95th percentiles of the distribution of CBO's projections of sea-level rise, respectively, given 3°C of global warming.

The Pacific Northwest is expected to see the smallest increase in the contiguous United States, with sea levels rising by 1.0 feet to 2.5 feet in the same scenario.

Natural Disasters

Researchers expect that higher sea levels, more intense storm surges, and heavier rainfall will make floods more frequent and extreme. By 2050, flood damage is projected to increase by one-quarter to one-third solely as a result of climate change.¹⁵

Climate change will also create hotter, drier conditions in some parts of the United States that are vulnerable to wildfires. The amount of land burned by wildfires has doubled over the past 30 years.¹⁶ Increases in temperatures dry out vegetation, making fires more likely to start and spread. Higher temperatures will probably also induce more ignitions by prompting more lightning strikes. However, predicting future wildfire risk is difficult because people play an important role in both starting and stopping fires. Unlike flood models, models that attempt to predict increases in wildfires due to climate change have not performed well.¹⁷

Uncertainty and Tipping Points

Many processes in the climate system are sufficiently well understood that associated risks can be confidently projected with clearly defined probability ranges. For other processes, however, the scientific community has not reached a consensus on the underlying mechanisms, or the data needed to make predictions are incomplete, inconsistent, or unreliable—in large part because projected levels of warming have no precedent in recorded history. In such cases, projections involve uncertainties that cannot be well quantified, and as a result, they are reported with low confidence and wide ranges of possible outcomes.

Warming within the range projected through the rest of the century might push one or more components of the climate system past critical thresholds—sometimes called tipping points—that would lead those components to shift to a very different state. (To conceptualize such a change in state, consider how warming ice remains solid until it reaches the threshold of melting, after which it becomes water.) However, those critical thresholds cannot yet be identified with confidence.

Research suggests that three types of climate processes would have especially large impacts if they reached such critical thresholds (see Figure 1-7):

- Greenhouse gas and temperature feedbacks would cause the release of substantially more greenhouse gases, thereby enhancing the amount and rate of warming.
- Changes to ocean circulation and weather patterns would shift regional temperatures and precipitation.
- The melting of ice sheets would dramatically change the amount and rate of sea-level rise.

Although some of those components of the climate system could reach critical thresholds by the end of the century, many of the effects would unfold over a longer time frame. Some of those effects would occur over the course of decades; others would unfold over centuries or even millennia. And some changes would be irreversible for hundreds or thousands of years, even if climate conditions returned to their present state sooner. Furthermore, crossing one critical threshold would probably change the likelihood of crossing others.

There is some chance that one or more thresholds have already been crossed, and that chance will generally increase with more warming (see Figure 1-8). But those critical thresholds are uncertain, and estimates of the size and likelihood of their effects will be revised as scientific understanding improves.

Greenhouse Gas and Temperature Feedbacks

Some of the Earth's natural features, such as permafrost and the Amazon rainforest, respond to warming by releasing additional amounts of greenhouse gases. Those additional greenhouse gases, in turn, increase the amount and pace of warming.

Permafrost. Permafrost—a mix of perennially frozen soil and rock—can extend from a few feet below the

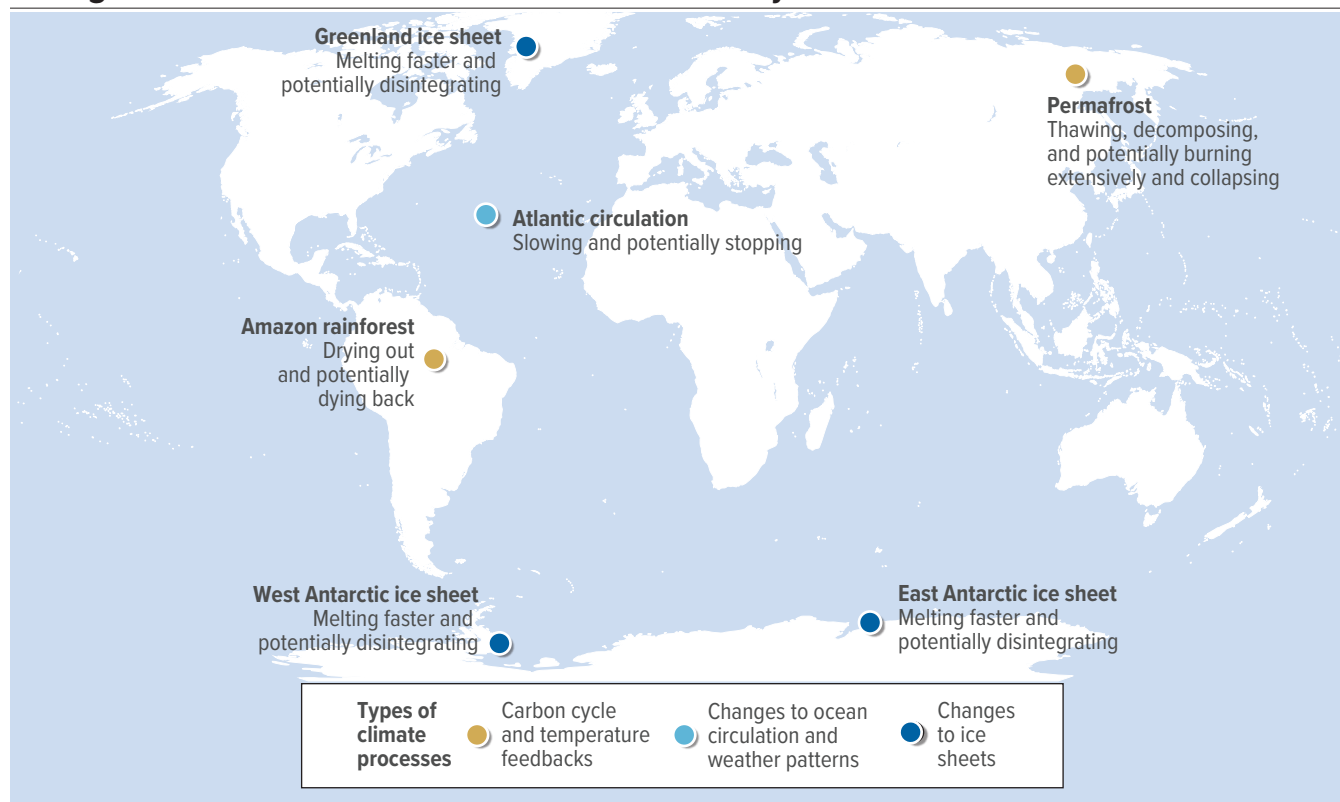
15. Congressional Budget Office, *Flood Damage and Federally Backed Mortgages in a Changing Climate* (November 2023), www.cbo.gov/publication/59379; and Oliver E. J. Wing and others, "Inequitable Patterns of U.S. Flood Risk in the Anthropocene," *Nature Climate Change*, vol. 12, no. 2 (February 2022), pp. 156–162, <https://doi.org/10.1038/s41558-021-01265-6>.

16. Congressional Budget Office, *Wildfires* (June 2022), www.cbo.gov/publication/57970.

17. For a discussion of wildfire modeling by insurers, see Congressional Budget Office, *Climate Change, Disaster Risk, and Homeowner's Insurance* (August 2024), www.cbo.gov/publication/59918.

Figure 1-7.

Changes in Natural Processes That Would Have Major Effects on the Climate



Adapted from Timothy M. Lenton and others, "Climate Tipping Points—Too Risky to Bet Against," *Nature*, vol. 575 (November 27, 2019), pp. 592–595, <https://doi.org/10.1038/d41586-019-03595-0>.

Earth's surface to nearly a mile deep. Organic matter in permafrost holds nearly twice as much carbon as the atmosphere does. Much of that carbon is in the form of methane, which traps heat more powerfully than CO₂. As permafrost thaws, its organic matter decomposes and releases both CO₂ and methane into the atmosphere. Permafrost has already begun to thaw gradually and to release greenhouse gases. As global temperatures continue to rise, that process will persist, causing modest increases in warming.¹⁸

Once warming reaches a critical threshold—somewhere between 1.0°C and 2.3°C—pockets of permafrost in many regions will thaw more abruptly.¹⁹ Over the course

of a few centuries, such regional thawing would increase emissions by 50 percent more than gradual thawing alone.

With even more warming, more extensive areas of permafrost would thaw, and some of the organic matter in that permafrost would dry out and become susceptible to fire. The heat generated by such burning could become self-sustaining, leading to the collapse of large regions of permafrost. Widespread burning of permafrost would release hundreds of gigatons of CO₂ over several decades or centuries, yielding roughly 0.2°C to 0.4°C of additional warming.²⁰ For such widespread burning to occur, global temperatures would probably have to rise by 3°C to 6°C.

Some permafrost lies underwater in shallow ocean sediments. In that permafrost are methane hydrates—mixtures of ice and methane. If exposed to enough warming, methane hydrates would be an extremely large source of emissions. However, researchers believe that only a small

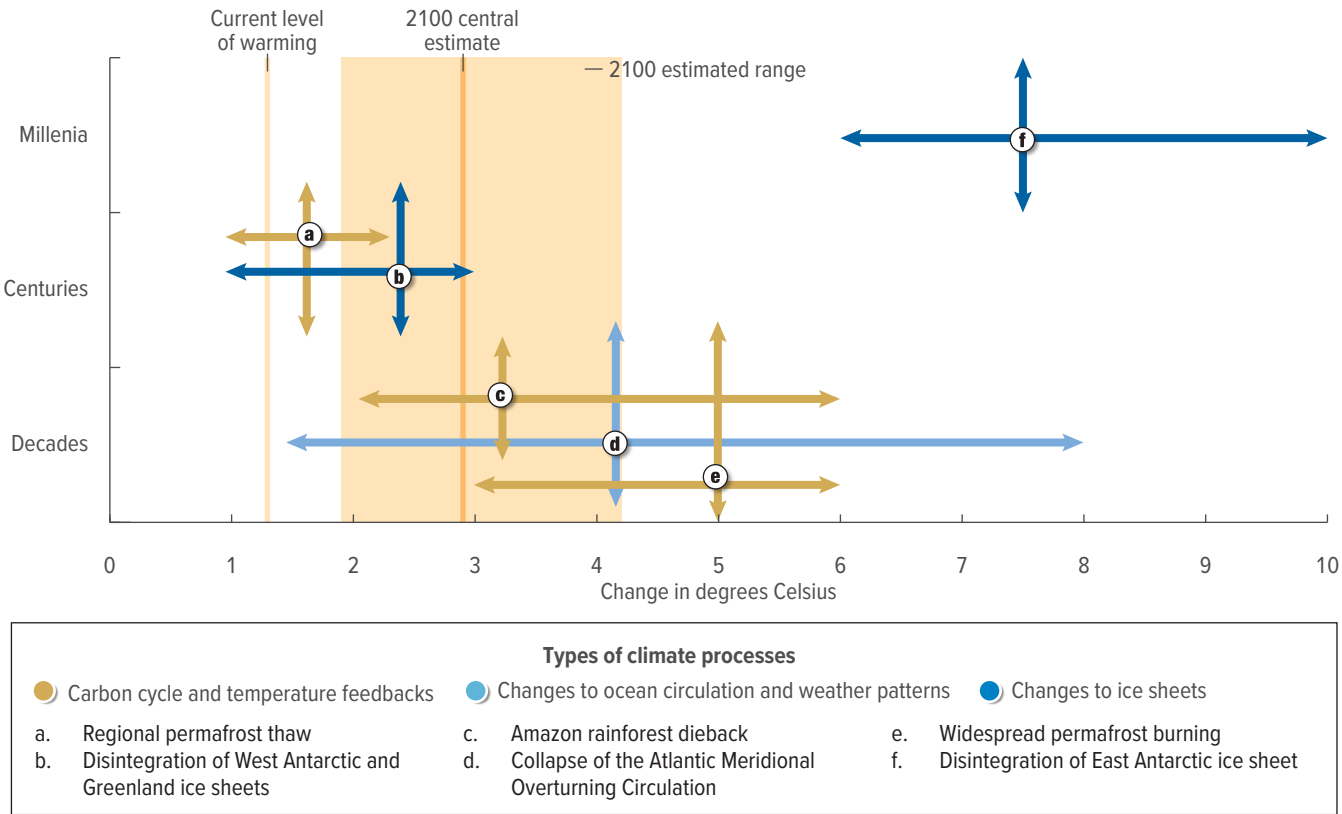
18. Seaver Wang and others, "Mechanisms and Impacts of Earth System Tipping Elements," *Reviews of Geophysics*, vol. 61, no. 1 (March 2023), <https://doi.org/10.1029/2021RG000757>.

19. David I. Armstrong McKay and others, "Exceeding 1.5°C Global Warming Could Trigger Multiple Climate Tipping Points," *Science*, vol. 377, no. 6611 (September 9, 2022), article eabn7950, <https://doi.org/10.1126/science.abn7950>.

20. Ibid. Those amounts are highly uncertain, in part because of uncertainty about what would grow in place of the permafrost and how much carbon it would absorb.

Figure 1-8.

Ranges of Temperature Thresholds and Time Frames for Major Changes in Climate Processes



Data source: Congressional Budget Office, using data from Timothy M. Lenton and others, *Global Tipping Points: Report 2023* (University of Exeter, 2023), <https://report-2023.global-tipping-points.org>; Seaver Wang and others, “Mechanisms and Impacts of Earth System Tipping Elements,” *Reviews of Geophysics*, vol. 61, no. 1 (March 2023), article e2021RG000757, <https://doi.org/10.1029/2021RG000757>; and David I. Armstrong McKay and others, “Exceeding 1.5°C Global Warming Could Trigger Multiple Climate Tipping Points,” *Science*, vol. 377, no. 6611 (September 9, 2022), article eabn7950, <https://doi.org/10.1126/science.abn7950>. See www.cbo.gov/publication/60845#data.

The temperature ranges and time periods presented in this figure are approximate.

portion of the methane in those deposits will be released over the coming century.²¹

The Amazon Rainforest. Warming is projected to make the Amazon rainforest drier and more vulnerable to drought, wildfires, and other threats. Parts of the

rainforest are sustained throughout its annual dry season by rainfall that is retained from its wet season. Warming is projected to disrupt that water cycle by reducing rainfall during the wet season and lengthening the dry season. Sufficient warming would lead to a self-sustaining process whereby much of the remaining forest dies and is replaced by drier savannah, which stores far less carbon. The retreat of the rainforest, referred to as dieback, would add on net up to 200 Gt of CO₂ to the atmosphere—the equivalent of about five years’ worth of current global CO₂ emissions from fossil fuel combustion and cement production. That process would probably occur over the course of several decades to a couple of centuries.

21. Josep G. Canadell and others, “Global Carbon and Other Biogeochemical Cycles and Feedbacks,” in Valérie Masson-Delmotte and others, eds., *Climate Change 2021: The Physical Science Basis, Working Group I Contribution to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* (Cambridge University Press, 2021), pp. 673–816, <https://doi.org/10.1017/9781009157896>.

However, the amount of warming needed to initiate that process is highly uncertain. In the absence of further deforestation, dieback would likely be initiated somewhere in the range of 2°C to 6°C of warming.²² The extent to which any given amount of deforestation would lower that threshold is unknown.

According to recent research, patterns of rainfall in the Amazon region will also be affected by changes in the Atlantic Meridional Overturning Circulation (AMOC), which is discussed further below.²³ The potential for such interactions among components of the climate system highlights the complexity of projecting the effects of warming on all of them together.

Changes to Ocean Circulation and Weather Patterns

Several components of the global ocean circulation system have powerful effects on regional temperatures and patterns of precipitation. The most important is the Atlantic Meridional Overturning Circulation, which includes the Gulf Stream. The AMOC carries warm water from the South Atlantic and the tropics to the North Atlantic. When it reaches high latitudes, the water evaporates and cools, becoming saltier and therefore denser, and sinks into the deep ocean. That downwelling helps drive a system of circulation that flows through the deep Atlantic and ultimately into the Indian and Pacific Oceans, playing a central role in maintaining ocean circulation patterns and climates around the world. If climate conditions slow or disrupt that downwelling (such as by changing the temperature or salinity of the water in the North Atlantic), the AMOC would slow down or even collapse—that is, stop—completely.

The AMOC collapsed repeatedly in the distant past. In the most recent instance, about 13,000 years ago, warming surface waters and melting ice contributed to a collapse of the AMOC as the Earth was emerging from the last ice age. That collapse led to a southward shift in the

tropical rain belt, weaker African and Asian monsoons in the Northern Hemisphere, stronger monsoons in the Southern Hemisphere, and cooler and drier conditions in Europe. Once the system collapses, it takes centuries to start circulating again.

The AMOC has weakened in recent decades and is expected to weaken further. However, it is unclear how much of that weakening is due to natural variability and how much is due to recent warming. As a result, future trends in the AMOC are highly uncertain. Collapse of the AMOC could be caused by as little as 1.4°C or as much as 8°C of warming.²⁴ It is therefore possible that nearly enough warming has already occurred to cause an eventual collapse of the AMOC, although the Intergovernmental Panel on Climate Change has stated that a collapse is unlikely to occur during this century.²⁵

Changes to Ice Sheets

A third group of potentially large impacts involves ice sheets—continent-sized bodies of ice covering land in Greenland and Antarctica. Ice sheets are responding to recent warming by melting at an accelerating rate. So far, only a small portion of the ice in the sheets (a fraction of a percent) has melted. It is uncertain how much mass the ice sheets will lose or how rapidly they will lose it in response to any increase in global temperatures. The most important uncertainties involve the effects of warming ocean waters on the bases of ice sheets below sea level. As discussed above, given the potential for rapid ice sheet loss due to uncertain processes, there is a risk that global sea-level rise will reach 5 feet or more by 2100.²⁶

With sufficient warming, each ice sheet will reach a critical threshold that causes its complete, irreversible disintegration. Far larger increases in sea levels would

22. Timothy M. Lenton and others, *Global Tipping Points: Report 2023* (University of Exeter, 2023), <https://report-2023.global-tipping-points.org>; and David I. Armstrong McKay and others, “Exceeding 1.5°C Global Warming Could Trigger Multiple Climate Tipping Points,” *Science*, vol. 377, no. 6611 (September 9, 2022), article eabn7950, <https://doi.org/10.1126/science.abn7950>.

23. Nico Wunderling and others, “Climate Tipping Point Interactions and Cascades: A Review,” *Earth System Dynamics*, vol. 15, no. 1 (January 2024), pp. 41–74, <https://doi.org/10.5194/esd-15-41-2024>.

24. David I. Armstrong McKay and others, “Exceeding 1.5°C Global Warming Could Trigger Multiple Climate Tipping Points,” *Science*, vol. 377, no. 6611 (September 9, 2022), article eabn7950, <https://doi.org/10.1126/science.abn7950>.

25. Hervé Douville and others, “Water Cycle Changes,” in Valérie Masson-Delmotte and others, eds., *Climate Change 2021: The Physical Science Basis, Working Group I Contribution to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* (Cambridge University Press, 2021), pp. 1055–1210, <https://doi.org/10.1017/9781009157896>.

26. William V. Sweet and others, *Global and Regional Sea Level Rise Scenarios for the United States: Updated Mean Projections and Extreme Water Level Probabilities Along U.S. Coastlines*, Technical Report NOS 01 (National Oceanic and Atmospheric Administration, February 2022), <https://tinyurl.com/cmsv48xr>.

follow. If the ice sheets disintegrated completely in Greenland and the western part of Antarctica, sea levels would rise by roughly 30 feet over centuries to millennia. That disintegration could be initiated by as little as 1°C to 3°C of warming.²⁷ In the very large eastern part of the Antarctic ice sheet, portions with bases below sea level could begin to melt with 2°C to 6°C of warming. The critical threshold for complete disintegration lies between 6°C and 10°C of warming—well outside the range of projections for the next century. The total loss of all ice

sheets would raise sea levels by roughly 200 feet, but the full effect would take thousands or tens of thousands of years to realize.

However, like projections of the amount and pace of ice loss, projections of the amount of warming needed to trigger the ice sheets' disintegration are highly uncertain. Because researchers cannot yet fully characterize and quantify the underlying processes, the Intergovernmental Panel on Climate Change has expressed a low degree of confidence about those projections.

27. Timothy M. Lenton and others, eds., *Global Tipping Points: Report 2023* (University of Exeter, 2023), <https://report-2023.global-tipping-points.org>.

Chapter 2: Effects on Economic Output

The Congressional Budget Office projects that climate change is likely to reduce the United States's output of goods and services during the 21st century. Climate change will affect output through its effects on the supply of labor (both the number of people who are working and the number of hours they work), productivity, and the stock of productive capital (also called the capital stock—machines, structures, and intellectual property used to produce goods and services).

Because those effects are uncertain, the overall effect of climate change on output is also uncertain. However, the distribution of possible outcomes shows that climate change is much more likely to decrease gross domestic product—the most commonly used measure of economic output—than to increase it.

How Climate Change Affects Output

Climate change will have a variety of effects on the main determinants of economic output—the supply of labor, productivity, and the capital stock. Efforts to prevent or mitigate damage from climate change will moderate those effects.

The Supply of Labor

At different times and in different places, climate change will cause increases and decreases in the number of lost workdays. More workdays will be lost because of high temperatures; disasters such as floods, wildfires, and hurricanes; and illnesses related to heat exposure, air pollution, and vector-borne diseases (those transmitted by mosquitoes and other organisms). Fewer workdays will be lost because of cold temperatures, winter storms, and cold-related illnesses and accidents. On the whole, CBO expects that the increases in lost workdays will be larger than the decreases. As a result, the supply of labor will be smaller than it would have been in the absence of additional climate change.

Productivity

For similar reasons, labor productivity—that is, output per hour worked—is expected to decline overall. In CBO's estimation, reductions in the productivity of construction workers, agricultural workers, and others who

work outside during hot weather will exceed increases in the productivity of outdoor workers in some locations during winter months.

The productivity of agricultural land is likewise expected to decline overall. In some areas, climate change will lead to warmer temperatures and longer growing seasons that boost agricultural production. However, climate change will also lead to increases in extreme temperatures, worse droughts and pressures on water supplies, and a greater prevalence of crop pests and diseases.

Finally, the private sector's production costs are expected to increase, on net, which will reduce output per hour worked. In certain areas and at certain times, heating costs will be lower than they would have been otherwise. But those savings will be outstripped by increases in the costs of air conditioning, temperature control in industries such as pharmaceutical manufacturing and goods transport, and damage to infrastructure.

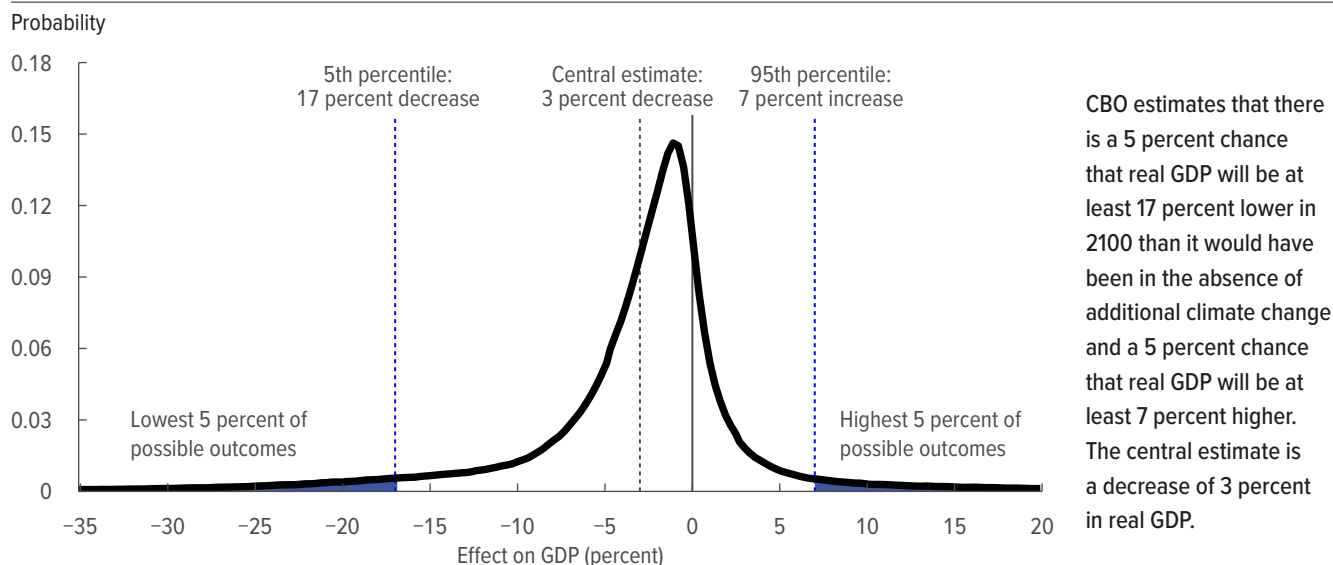
The Capital Stock

Natural disasters and other aspects of climate change are expected to reduce the capital stock—and therefore GDP—through several channels. In the short run, factories that lose productive capital will no longer be able to produce the same amount of output, and GDP will fall to the extent that other U.S. factories cannot make up the difference. If the lost capital is replaced, those investments will contribute to GDP in the year the purchases are made. Nevertheless, the economy's capital stock will be smaller than it would have been otherwise—either because capital that was destroyed is not replaced or because replacing it diverts funds from investments in new productive capital. With a smaller capital stock, production will be lower than it would have been otherwise, and so will GDP.

In the long run, greater risks of losses to productive capital will reduce the expected returns on investments that motivate investors' decisions to invest. As a result, investors will invest less overall, further diminishing the capital stock and, in turn, GDP.

Figure 2-1.

The Distribution of Possible Changes to Real GDP in 2100 From Climate Change



Data source: Congressional Budget Office. See www.cbo.gov/publication/60845#data.

Real values are nominal values that have been adjusted to remove the effects of inflation.

GDP = gross domestic product.

Adaptation

People, businesses, and governments will adapt to climate change by altering their behavior, investing in buildings and equipment that are more appropriate to the changing climate, and developing new technologies to mitigate its impact. On the whole, such adaptations are expected to reduce the impact of climate change on GDP. (Some adaptations, such as people's relocations from regions of the country that have become uncomfortably hot to regions that have become more temperate, would involve personal losses of well-being that are not counted in GDP.)

The economic effects of investments in adaptations involving buildings and equipment (such as air conditioning, irrigation systems, or levees) are complex. Like investments to replace destroyed or damaged capital, those investments will boost GDP in the year that the spending occurs but will crowd out some other investments that, in the absence of climate change, would have been more productive.

Projected Effects on GDP

By the end of the 21st century, in the absence of further climate change, real GDP is projected to be more than three times what it was in 2024. CBO estimated the

possible effects of climate change on GDP by 2100. The distribution of effects suggests that climate change is much more likely to reduce that growth in GDP than to increase it.

In CBO's estimation, there is a 5 percent chance that real (inflation-adjusted) GDP will be at least 17 percent lower in 2100 than it would have been in the absence of additional climate change and a 5 percent chance that it will be at least 7 percent higher (see Figure 2-1). In 2023, changes in GDP of that size would have amounted to a \$4.7 trillion decrease and a \$1.9 trillion increase. The agency's central estimate is that climate change will cause real GDP to be 3 percent lower than it would have been if temperatures remained stable after 2024. Climate change is expected to have a larger negative effect on GDP in some other countries, particularly those that are less developed or have a hotter climate.

Uncertainty Surrounding the Effects of Tipping Points

The estimates of the effects of climate change on GDP described above are based on the historical relationship between temperature and the production of goods and services. They therefore do not account for the effects of potential developments with no recorded historical

Box 2-1.**Limitations of GDP as a Measure of Economic Activity**

In this analysis, the Congressional Budget Office follows the common practice of estimating the effects of climate change on gross domestic product (GDP) rather than on some other measure of economic activity. Despite its widespread use, GDP has shortcomings as a measure of economic activity and the economic effects of climate change.

Economic Activity in Other Countries

GDP measures only the value of goods and services produced within the borders of a country—it excludes the value of economic activity in other countries. Because Americans benefit from trade with other countries, they will be affected by the economic effects of climate change not only in the United States but around the world. In 2023, U.S. exports totaled about \$3 trillion, or 11 percent of GDP, and U.S. imports totaled nearly \$4 trillion, or 14 percent of GDP.¹

In addition, at the end of the second quarter of 2024, Americans owned \$36 trillion in foreign assets.² Impacts on those

assets would not be captured in GDP. Instead, they would be captured in gross national product, which measures the income of Americans regardless of where that income is earned.

Capital Depreciation

Other issues concern how GDP overstates the nation's ability to produce goods and services on an ongoing basis. GDP includes investments that simply replace depreciated capital—that is, capital that has been used up in the process of production by wear and tear, obsolescence, or normal accidental damage. An alternative measure of output, net domestic product, excludes those investments, thus providing a more accurate picture of output after maintaining the capital stock. In addition, GDP does not account for the loss in productive capacity when naturally occurring environmental resources such as clean air, rivers, and wetlands are damaged or used up.

A related issue is that GDP does not directly reflect damage to capital from unusual events such as natural disasters, even if the capital can no longer be used in production. As a result, when damage from hurricanes, floods, or wildfires reduces the capital stock, GDP does not change. However, investments to replace the lost capital are counted in GDP. The resulting increase in GDP thus overstates the extent to which those investments improve the economy.

1. Bureau of Economic Analysis, "National Income and Product Accounts Table 1.1.5. Gross Domestic Product" (accessed October 28, 2024), <https://tinyurl.com/pckf33tj>.

2. Bureau of Economic Analysis, "U.S. International Investment Position, 2nd Quarter 2024," BEA-24-26 (news release, September 25, 2024), <https://tinyurl.com/3b6c35fs>.

precedent (see Chapter 1). Examples of such developments include greater warming in response to the release of natural stores of greenhouse gases, greater sea-level rise in response to more rapid melting of ice sheets in Greenland and Antarctica, and shifts in weather patterns driven by disruptions in ocean circulation patterns.

Given the great uncertainty surrounding those developments and the thresholds at which they would occur (sometimes called tipping points), research on their economic consequences is limited. Their unprecedented nature makes it difficult to anticipate all of their effects or whether those effects will interact. Because it is impossible to fully account for the risks of such uncertain changes, the true extent of the damage from climate change (and the benefits from mitigating it) has the potential to be much greater than the estimates presented here.¹

1. For one of the few studies of the economic effects of climate change that has accounted for several (though not all) potential interactions among tipping points, see Simon Dietz and others,

CBO's Analytical Approach

CBO developed its projections by drawing on 14 studies published between 2016 and 2024, each of which provided estimates of the effect of temperature change on GDP for the United States or, in a few instances, North America.² (For a discussion of some limitations of GDP as a measure of economic activity, see Box 2-1.) Changes in average temperature serve as a proxy for changes in precipitation and wind patterns and for daily and seasonal temperature extremes; incorporating those other measures usually has only a small effect on estimates of

"Economic Impacts of Tipping Points in the Climate System," *Proceedings of the National Academy of Sciences of the United States of America*, vol. 118, no. 34 (August 16, 2021), article e2103081118, <https://doi.org/10.1073/pnas.2103081118>. The authors of that study concluded that interactions among tipping points only modestly increased their estimated economic impact.

2. See "Estimates of the Effect of Temperature Change on GDP" in the appendix.

GDP once changes in average temperature are taken into consideration.³

The studies took different approaches to estimate the effect of climate change on GDP. The differences concerned three characteristics:

- The geographic scope of the analysis. Some studies estimated the effect on the basis of experiences in the United States alone. Others estimated the effect on the basis of experiences around the world.
- The method used. Some studies constructed estimates by summing the effects of climate change through specific channels, such as labor productivity, agricultural output, and damage to coastal properties caused by disasters. Others directly estimated the relationship between climate change and GDP.
- Persistence of the effect. Some studies considered the effect of climate change on GDP to be short-lived. Others considered the effect to persist for several years or to be permanent.

To synthesize the results from those studies, CBO conducted a meta-analysis, which involved calculating a weighted average of their results. CBO used the foregoing characteristics to assign weights to the studies that reflected their relevance to the effects of climate change

on the United States. For instance, CBO assigned more weight to studies that focused exclusively on the United States than to studies that also incorporated the effects of climate change on other countries that will experience different physical effects or have different economic systems. CBO also adjusted the results to account for the effects of adaptations to climate change.⁴

CBO's analysis differs from other meta-analyses in the research literature in three ways. First, the agency excluded studies published before 2016 and studies that estimated effects other than those on GDP. Second, the agency adopted its own criteria for weighting studies. And third, the agency estimated a distribution of the effect of climate change on GDP that is not tied to any single temperature outcome.

An alternative to CBO's approach appears in a series of reports produced by the Council of Economic Advisers and the Office of Management and Budget. The authors of those reports analyzed the effects of climate change on the U.S. economy 10 years or 25 years into the future. In that work, they equally weighted estimates from the studies included in their analyses to provide a central estimate.⁵

3. Studies may attribute small effects to other aspects of climate change because the effects actually are small, because they are small for the conditions typically experienced, or—in the case of some aspects of climate change, such as precipitation—because the effects are important primarily at a certain time of year or in a certain area but are averaged across an entire year or over large geographic areas in the data. For one study that found an independent effect of precipitation on GDP in U.S. counties, see Kamiar Mohaddes and others, “Climate Change and Economic Activity: Evidence From U.S. States,” *Oxford Open Economics*, vol. 2 (2023), article odac010, <https://doi.org/10.1093/ooec/odac010>.

4. For more details about CBO's methods and the studies on which the agency's analysis is based, see Chad Shirley and William Swanson, *The Effects of Climate Change on GDP in the 21st Century* (Congressional Budget Office, forthcoming).

5. Council of Economic Advisers and Office of Management and Budget, *Assessing Methods to Integrate the Physical Risks and Transition Risks and Opportunities of Climate Change Into the President's Macroeconomic Forecast* (April 2024), <https://tinyurl.com/wbkeprh8>, *Methodologies and Considerations for Integrating the Physical and Transition Risks of Climate Change Into Macroeconomic Forecasting for the President's Budget* (March 2023), <https://tinyurl.com/3zek6v33>, and *Climate-Related Macroeconomic Risks and Opportunities* (April 2022), <https://tinyurl.com/4v3nnns9>.

Chapter 3: Effects on Real Estate and Financial Markets

The risk of climate-induced property damage will increase throughout the 21st century. Much of that damage will be due to rising sea levels, which will put some coastal properties permanently underwater. Increases in precipitation in many parts of the United States will also raise the risk of flooding. Hotter and drier conditions in other parts of the country will increase the length of fire seasons and the intensity of wildfires, putting more real estate at risk. All of those risks will be exacerbated by continued development in areas that are vulnerable to flooding and wildfires. Given that increased risk of damage, insurance rates are likely to rise, and property values are likely to be lower than they would be without further climate change.

Those losses in property values will be borne by some combination of property owners, mortgage lenders, insurance companies, and the federal government. If the losses were large enough and widespread enough, they would lead to cascading failures of financial institutions that would undermine the financial system. Although the Congressional Budget Office cannot determine the likelihood of that risk, it would probably depend on the amount of future losses, how quickly they were realized, and how concentrated they were in certain areas or among certain lenders.

Property at Risk

Climate change will put properties at greater risk of damage from floods, hurricanes, and wildfires. The extent of future property damage will depend not only on climate change but on population growth and development in risky areas and investments in adaptive measures.

Damage From Floods

Climate change will affect flood damage to inland and coastal properties in different ways. About 17 million properties currently face at least a 1 percent chance of flooding each year; of those, roughly 13 million are inland properties and roughly 4 million are coastal properties.¹ In inland areas, flood damage will increase

because a warmer atmosphere holds more moisture, resulting in heavier rainfall from hurricanes and storms. For some coastal properties, sea-level rise will lead to tidal flooding—that is, regular flooding during high tides.² Still more coastal properties will face increasingly damaging storm surges as sea levels rise and warmer ocean temperatures cause more intense hurricanes. In CBO’s estimation, a growing share of damage from flooding will occur in coastal areas, including two-thirds of the damage to homes with mortgages in 2050.³

By 2100, rising sea levels will put a substantial amount of coastal real estate at risk of tidal flooding. However, estimates of that amount vary. One study estimated that if sea levels rose by an average of approximately 4 feet nationally and no further adaptations were implemented, about \$930 billion in residential property (about 2 percent of the current value of U.S. residential real estate) would be flooded at least twice a month by 2100. With approximately 2 feet of sea-level rise, the value of property at risk would be about \$250 billion (0.5 percent of the current value of residential real estate).⁴ Even residential property that is not routinely flooded will likely see its value drop if tidal flooding affects local businesses and infrastructure.⁵

1. Congressional Budget Office, *Communities at Risk of Flooding* (September 2023), www.cbo.gov/publication/58953.

2. Sea levels will not rise uniformly across the United States. For an illustration of projected regional increases in sea levels, see Figure 1-6 on page 12.

3. Congressional Budget Office, *Flood Damage and Federally Backed Mortgages in a Changing Climate* (November 2023), www.cbo.gov/publication/59379.

4. Union of Concerned Scientists, *Underwater: Rising Seas, Chronic Floods, and the Implications for U.S. Coastal Real Estate* (June 2018), www.ucsusa.org/resources/underwater. Property values were adjusted for growth from 2016 to 2023 using data from Treh Manhertz, “The U.S. Housing Market Gained More Value in 2020 Than in Any Year Since 2005” (Zillow, January 26, 2021), <https://tinyurl.com/5vmdsp86>; and Orphe Divounguy, “The Value of Residential Real Estate Broke a New Record \$52 Trillion” (Zillow, September 26, 2023), <https://tinyurl.com/59c8yhcx>.

5. T. M. Logan, M. J. Anderson, and A. C. Reilly, “Risk of Isolation Increases the Expected Burden From Sea-Level Rise,” *Nature Climate Change*, vol. 13, no. 4 (April 2023), pp. 397–402, <https://doi.org/10.1038/s41558-023-01642-3>.

Effects of Coastal Development. Estimates of the value of property at risk of flooding are based on current patterns of real estate development. Damages will be greater if the amount of people and property in coastal areas continues to grow. Population growth in coastal counties is projected to roughly double the number of U.S. residents living in homes at risk of inundation by 2100.⁶ CBO's analysis suggests that coastal development will increase the costs of flood damage as much as or more than climate change alone.⁷

Effects of Adaptations. Adaptation could mitigate much of the expected increase in property damage due to climate change. Adaptations that protect against rising sea levels include elevating homes, replenishing the sand on beaches, and building seawalls. For inland properties, adaptations include dredging rivers and building dams and levees. Some studies have estimated that adaptations have the potential to reduce future flood damage by large amounts—from 50 percent to 75 percent, depending on whether the adaptations are built to withstand future increases in flood risk from sea-level rise.⁸ In practice, damages will likely reflect an outcome in between a scenario with no further adaptations and a scenario in which people perfectly anticipate future increases in flood risk and invest in adaptations accordingly.

How much action will be taken in advance to adapt to the effects of climate change is uncertain. Many adaptations that would cost-effectively protect against current threats, irrespective of climate change, have not been adopted.⁹ Communities tend to implement adaptations after, rather than before, a disaster occurs.¹⁰ Most

adaptations involve substantial costs, which discourages their implementation, and they often require community coordination or approval. Moreover, even though adaptations reduce the risk of flood damage for existing structures, they can also create incentives to continue building in risky areas, putting more property at risk of damage.¹¹

Planning future adaptations poses challenges because it is difficult for most people to assess their exposure to changing climate risks. Federal Emergency Management Agency (FEMA) flood maps are meant to indicate current levels of flood risk, but they are updated infrequently and provide no information about potential future increases in risk. Even when communities can obtain detailed risk information, the uncertainty surrounding climate projections makes it difficult for them to evaluate the cost-effectiveness of protective measures.

Damage From Wildfires

Losses from wildfires have rapidly increased in recent years as fires have become larger and development has expanded into exposed areas.¹² Several factors have contributed to the increase in wildfires' size and severity. Much of the recent increase in the size of wildfires has been attributed to higher temperatures and increased dryness from climate change.¹³ In addition, forest management practices for much of the past century have allowed vegetation to grow denser, leading to more severe fires. And more than half of single-family homes built since 2010 are within or near wildlands, which increases the risk of human-caused fires and property damage.¹⁴ Roughly 2 million properties in the contiguous United States face at least a 1 percent chance of

6. Mathew E. Hauer, Jason M. Evans, and Deepak R. Mishra, "Millions Projected to be at Risk From Sea-Level Rise in the Continental United States," *Nature Climate Change*, vol. 6, no. 7 (July 2016), pp. 691–695, <https://doi.org/10.1038/nclimate2961>.
7. Congressional Budget Office, *Potential Increases in Hurricane Damage in the United States: Implications for the Federal Budget* (June 2016), www.cbo.gov/publication/51518.
8. Environmental Protection Agency, *Technical Documentation for the Framework for Evaluating Damages and Impacts (FrEDI)*, EPA 430-R-21-004 (August 2024), <https://tinyurl.com/bdhnbnw3>.
9. Mark Lorie and others, "Modeling Coastal Flood Risk and Adaptation Response Under Future Climate Conditions," *Climate Risk Management*, vol. 29 (April 2020), article 100233, <https://doi.org/10.1016/j.crm.2020.100233>.
10. Yanjun (Penny) Liao, Simon Sølvsten, and Zachary Whitlock, *Community Responses to Flooding in Risk Mitigation Actions: Evidence From the Community Rating System*, Working Paper 24-08 (Resources for the Future, June 2024), <https://tinyurl.com/yd46nyzk>.

11. Morgan J. Breen, Abiy S. Kebede, and Carola S. König, "The Safe Development Paradox in Flood Risk Management: A Critical Review," *Sustainability*, vol. 14, no. 24 (December 17, 2022), article 16955, <https://doi.org/10.3390/su142416955>.
12. Congressional Budget Office, *Wildfires* (June 2022), www.cbo.gov/publication/57970.
13. Marco Turco and others, "Anthropogenic Climate Change Impacts Exacerbate Summer Forest Fires in California," *Proceedings of the National Academy of Sciences of the United States of America*, vol. 120, no. 25 (June 20, 2023), article e2213815120, <https://doi.org/10.1073/pnas.2213815120>; and John T. Abatzoglou and A. Park Williams, "Impact of Anthropogenic Climate Change on Wildfire Across Western U.S. Forests," *Proceedings of the National Academy of Sciences of the United States of America*, vol. 113, no. 42 (October 10, 2016), pp. 11770–11775, <https://doi.org/10.1073/pnas.1607171113>.
14. Lily Katz and Taylor Marr, "America Is Increasingly Building Homes in Disaster-Prone Areas," *Redfin News* (September 9, 2022), <https://tinyurl.com/2few6xsr>.

damage from wildfires each year.¹⁵ Over the past decade, an average of about 8,000 structures per year have been destroyed by wildfires.¹⁶

Estimates of future wildfire risk vary widely. From 2014 to 2023, the average annual cost of major wildfires to homeowners, businesses, and governments exceeded \$10 billion (in 2024 dollars).¹⁷ One study estimated that by 2100, the value of property at high risk of wildfire damage would increase 10-fold if warming reached 4°C and 5-fold if warming reached roughly 3°C.¹⁸ Those estimates, which were based on home prices in 2010, already understate the value of property at risk today due to development in wildfire-exposed areas. Adaptations such as clearing vegetation around homes, implementing better forest management practices, and requiring the use of fire-resistant materials as part of building codes would help attenuate those losses.¹⁹

Bearers of Financial Risks Associated With Property Damage

Different participants in the real estate market—including property owners, mortgage lenders, insurance providers, and the federal government—will bear losses because of climate change. Those participants all face the risk of losses from property damage caused by natural disasters and sea-level rise. Property owners and mortgage lenders also face the risk of declines in market prices due to the threat of that damage. The federal government takes on portions of both kinds of financial risk in different ways.

Inundation due to sea-level rise is expected to be the primary cause of real estate losses. At some point, the prices of homes at risk of inundation will fall. However, the extent to which prices for those homes will decline

before they experience regular flooding is uncertain. If a shift in homebuyers' perceptions of climate risk occurred as a result of a large natural disaster, that would decrease the demand for and prices of exposed real estate.²⁰ Widespread declines in real estate values are possible, depending on the pace of sea-level rise, the extent of adaptations to sea-level rise, and how demand for exposed real estate evolves before homes are inundated. If those declines in prices were large enough and widespread enough, they would lead to cascading failures of financial institutions that would undermine the entire financial system. CBO has not assessed the likelihood of those developments, which would probably depend on the amount of losses, how quickly they were realized, and how concentrated they were in certain areas or among certain lenders.²¹

The likelihood of downward repricing events will also depend on the extent to which current real estate prices already account for the risk of future damage. One recent study estimated that the total value of homes in the U.S. market would fall by \$187 billion if home prices accounted for exposure to all forms of flood risk through 2050.²² Little research has assessed overvaluation in markets for commercial real estate, but commercial buyers of residential real estate tend to demand larger discounts for exposed properties.²³

Home prices may not fully reflect future risks related to climate change for several reasons. Some homeowners are optimistic about their exposure to increases in sea levels (which in most cases will occur gradually, decades in the

15. First Street Foundation, *The 5th National Risk Assessment: Fueling the Flames* (May 2022), <https://tinyurl.com/47v6nhab>.

16. National Interagency Fire Center, "Wildland Fire Summaries" (accessed October 4, 2024), www.nifc.gov/fire-information/statistics.

17. That estimate includes only fires whose cost exceeded \$1 billion in 2024 dollars. See National Oceanic and Atmospheric Administration, "Billion-Dollar Weather and Climate Disasters" (accessed August 11, 2024), www.ncei.noaa.gov/access/billions.

18. William R. L. Anderegg and others, "Climate Change Greatly Escalates Forest Disturbance Risks to U.S. Property Values," *Environmental Research Letters*, vol. 18, no. 9 (September 2023), article 094011, <https://doi.org/10.1088/1748-9326/ace639>.

19. Jeffrey Czajkowski and others, *Application of Wildfire Mitigation to Insured Property Exposure*, CIPR Research Report (National Association of Insurance Commissioners, November 2020), www.rms.com/offer/wildfire-mitigation.

20. See Celso Brunetti and others, "Climate-Related Financial Stability Risks for the United States: Methods and Applications," *Economic Policy Review*, Federal Reserve Bank of New York, vol. 30, no. 1 (October 2024), <https://doi.org/10.59576/epr.30.1.1-37>; and Celso Brunetti and others, "Climate Change and Financial Stability," *FEDS Notes* (Board of Governors of the Federal Reserve System, March 19, 2021), <https://tinyurl.com/2xsnympn>.

21. For a discussion of the challenges of assessing climate risks in financial markets, see Board of Governors of the Federal Reserve System, *Pilot Climate Scenario Analysis Exercise: Summary of Participants' Risk-Management Practices and Estimates* (May 2024), <https://tinyurl.com/5dy7y9by>.

22. Jesse D. Gourevitch and others, "Unpriced Climate Risk and the Potential Consequences of Overvaluation in U.S. Housing Markets," *Nature Climate Change*, vol. 13 (March 2023), pp. 250–257, <https://doi.org/10.1038/s41558-023-01594-8>.

23. Rogier Holtermans, Dongxiao Niu, and Siqi Zheng, "Quantifying the Impacts of Climate Shocks in Commercial Real Estate Markets," *Journal of Regional Science*, vol. 64, no. 4 (June 2024), pp. 1099–1121, <https://doi.org/10.1111/jors.12715>.

future). Some expect that future adaptations will reduce flood risks, and others are unaware of the long-term threat. Subsidies and limits on premium increases also might delay reductions in prices for vulnerable residential properties because they encourage property owners and mortgage lenders to minimize or ignore flood risks.

Beyond potential impacts on financial institutions, losses to real estate would affect gross domestic product by reducing the residential services that people receive from having homes in which to live. Although the assets at risk are valued at \$930 billion, the value of those services and, therefore, the potential losses to GDP would be much smaller—amounting to a few tenths of a percent of GDP, in CBO’s estimation. Those losses would be small in relation to overall losses, and they are accounted for by some (though not all) of the reports CBO used to estimate GDP losses resulting from climate change.

Damage from climate change will have ramifications for financial markets, but so will policies implemented to reduce greenhouse gas emissions and the risk of climate change. Such policies would affect the economy or financial markets by reducing the value of certain assets or the profitability of certain businesses (see Box 3-1).

Property Owners

Homeowners leave themselves vulnerable to losses when they do not purchase insurance. Home prices appear to partially reflect *current* flood and wildfire risks. The extent to which they reflect *future* risks associated with climate change—particularly the risk of flooding from sea-level rise—is unclear.

Risks of Uninsured Losses. Many people with properties at risk of flooding do not have flood insurance. Standard homeowner’s insurance policies do not include it. However, many homeowners (more than half, according to one survey) wrongly believe that their homeowner’s policy covers flood damage.²⁴ Most people who purchase flood coverage do so because they are required to, either by their mortgage lender or by the federal government. Homes with federally backed mortgages in areas that face at least a 1 percent chance of flooding each year—areas designated by FEMA as Special Flood Hazard Areas (SFHAs) or 100-year floodplains—are

required to have flood insurance. But that mandate has not been complied with fully.²⁵ In addition, that requirement does not apply to homes that are owned outright. As a result, some homeowners do not purchase insurance, or they drop their policies when prices increase. In 2023, only about 20 percent of properties in SFHAs had flood insurance coverage.²⁶

In general, homeowners underestimate their flood risk.²⁷ Some may be unaware of their flood risk because SFHAs do not include all areas at risk of damage: They do not capture areas with an annual chance of flooding of less than 1 percent, nor do they account for all causes of flooding or future changes in risk. Many SFHAs also have outdated boundaries. CBO recently estimated that homes outside of SFHAs account for up to half of the flood damage to residential properties that is projected to occur each year.²⁸

Homeowners who are unaware of their risk or unwilling to pay an insurance rate that reflects it will probably draw on other sources to cover some costs in the wake of disasters, such as federal disaster assistance (though that assistance typically covers only a small portion of residential property losses).²⁹ In states with nonrecourse laws, borrowers who default on their mortgage are not liable for their unpaid mortgage debt. Some homeowners in those states may rely on their ability to default instead of purchasing disaster insurance, because they would lose

24. Center for Insurance Policy and Research, *Extreme Weather and Property Insurance: Consumer Views* (National Association of Insurance Commissioners, August 2021), <https://tinyurl.com/3kvea8jh>.

25. Government Accountability Office, *National Flood Insurance Program: Congress Should Consider Updating the Mandatory Purchase Requirement*, GAO-21-578 (July 30, 2021), www.gao.gov/products/gao-21-578.

26. Congressional Budget Office, *Flood Insurance in Communities at Risk of Flooding* (July 2024), www.cbo.gov/publication/60042.

27. For example, see Katherine R. H. Wagner, “Adaptation and Adverse Selection in Markets for Natural Disaster Insurance,” *American Economic Journal: Economic Policy*, vol. 14, no. 3 (August 2022), pp. 380–421, <https://doi.org/10.1257/pol.20200378>.

28. Congressional Budget Office, *Flood Damage and Federally Backed Mortgages in a Changing Climate* (November 2023), www.cbo.gov/publication/59379.

29. Carolyn Kousky, Erwann O. Michel-Kerjan, and Paul A. Raschky, “Does Federal Disaster Assistance Crowd Out Flood Insurance?” *Journal of Environmental Economics and Management*, vol. 87 (January 2018), pp. 150–164, <https://doi.org/10.1016/j.jeem.2017.05.010>.

only their equity in the case of a disaster.³⁰ Those homeowners will have more incentive to default in the future if flood events shift perceptions in the market of the long-term viability of coastal areas exposed to sea-level rise. Other homeowners will recover losses by selling their damaged home to a buyer who will tear it down and build a new one. Such sales are especially common in desirable locations when insurance payments would not cover the cost of rebuilding older homes to meet new building codes.

Risks to Property Prices From Tidal Flooding. Rising sea levels in the coming decades will put some homes at risk of inundation from tidal flooding. If sea levels rise enough and those homes are not protected by some form of adaptation, they will presumably lose all of their value. Exposed homes will lose value before they routinely experience flooding if buyers expect their resale value to fall in the future, when hazards escalate.

Evidence is mixed on the extent to which property values currently account for the risk of future tidal flooding. Some studies have found no discount for properties at risk of being inundated if sea levels rose by 6 feet or less.³¹ Other studies have observed discounts of 3 percent to 7 percent for such properties.³² One study in particular found larger discounts for properties in more imminent danger of sea-level rise: a 15 percent discount for homes that would suffer tidal flooding with a 1-foot increase in sea levels and an 8 percent discount for homes that would suffer such flooding with an increase of 4 feet to 5 feet.³³ Commercial buyers (who typically have more information about climate predictions than residential

property buyers) and residents of communities that are more concerned about climate change are more likely to pay prices that reflect future risks.³⁴

However, some discounts associated with sea-level rise seem to reflect only current flood risks, not the greater risks expected in the future. For example, in a study by Freddie Mac, homes exposed to rising sea levels sold at a discount only if they were in SFHAs, where flood insurance is required for a mortgage.³⁵ If the risk of flooding from sea-level rise is not reflected in prices now, homes are likely to lose value in the future as the risk becomes more apparent.

It is difficult to estimate how prices will change for homes that are not at risk of tidal flooding but are located in communities that will be broadly affected by sea-level rise. If some of the homes in a community become inundated, several effects are likely. To the extent that displaced residents compete for the remaining real estate, the value of safer homes will rise.³⁶ If displaced residents move away, reduced demand for housing will lower real estate prices in the area. Real estate prices will also fall if residents in nearby unaffected areas leave because of abandoned, unrepaired properties and damaged amenities and infrastructure. Any declines in property values will diminish the tax base and reduce funding for public services, thereby prompting more outmigration and reductions in home prices.³⁷

Risks to Property Prices From Other Types of Flooding. Some homes will be safe from inundation due to sea-level rise alone but will be exposed to increasingly dangerous storm surges or inland flooding. Properties

30. Philip Mulder and Yanjun Liao, *What's at Stake? Understanding the Role of Home Equity in Flood Insurance Demand*, Working Paper 24-06 (Department of the Treasury, Office of Financial Research, July 2024), <https://tinyurl.com/32cjze44>; and Laura Bakkensen, Toan Phan, and Tsz-Nga Wong, *Leveraging the Disagreement on Climate Change: Theory and Evidence*, Working Paper 23-01R (Federal Reserve Bank of Richmond, May 2024), <https://doi.org/10.21144/wp23-01>.

31. For example, see Justin Murfin and Matthew Spiegel, "Is the Risk of Sea Level Rise Capitalized in Residential Real Estate?" *Review of Financial Studies*, vol. 33, no. 3 (March 2020), pp. 1217–1255, <https://doi.org/10.1093/rfs/hhz134>.

32. See "Sea-Level Rise and Home Prices: State and National Studies" in the appendix.

33. Asaf Bernstein, Matthew T. Gustafson, and Ryan Lewis, "Disaster on the Horizon: The Price Effect of Sea Level Rise," *Journal of Financial Economics*, vol. 134, no. 2 (November 2019), pp. 253–272, <https://doi.org/10.1016/j.jfineco.2019.03.013>.

34. For example, see Markus Baldauf, Lorenzo Garlappi, and Constantine Yannelis, "Does Climate Change Affect Real Estate Prices? Only If You Believe In It," *Review of Financial Studies*, vol. 33, no. 3 (March 2020), pp. 1256–1295, <https://doi.org/10.1093/rfs/hhz073>.

35. Freddie Mac, *Sea Level Rise and Impact on Home Prices in Coastal Florida*, Economic and Housing Research Note (March 2022), <https://tinyurl.com/4usrfwrk>.

36. Jesse M. Keenan, Thomas Hill, and Anurag Gumber, "Climate Gentrification: From Theory to Empiricism in Miami-Dade County, Florida," *Environmental Research Letters*, vol. 13, no. 5 (May 2018), article 054001, <https://doi.org/10.1088/1748-9326/aabb32>.

37. Evelyn G. Shu and others, "Integrating Climate Change Induced Flood Risk Into Future Population Projections," *Nature Communications*, vol. 14 (December 2023), article 7870, <https://doi.org/10.1038/s41467-023-43493-8>.

Box 3-1.

Transition Risks

Transitioning to an economy with lower emissions will cut the risk of damage due to climate change and provide opportunities for new products and markets, but it will also impose economic and financial risks. The magnitude of those risks will largely depend on the stringency of emission-reduction policies. In the United States, the risks would mostly be borne by people who own, invest in, or work for private companies that hold extensive reserves of fossil fuels and specialized equipment for extracting and processing them. (In countries where substantial reserves are state-owned, the risks will largely be borne by the government.)

Stranded Assets Due to Fossil Fuel Reductions

If fully extracted and consumed, proven reserves of fossil fuels would produce well over 3,000 gigatons (Gt) of carbon dioxide (CO₂).¹ Policies to limit warming would entail abandoning a portion of those reserves (that is, leaving them in the ground); those policies would also reduce profits from refineries and other assets used to produce fossil fuels and from power plants that generate electricity by burning them. Fossil fuels

that continued to be sold would likely earn lower returns because of reduced demand. Some fossil fuel reserves and equipment would be written off (that is, their value would be reduced to zero for accounting purposes). Assets that suffer such losses are referred to as *stranded assets*.

The more stringent the policies, the greater the losses would be. According to one study, to have a 50 percent chance of limiting global warming to 2 degrees Celsius (2°C) or less, policies would have to limit global CO₂ emissions to an average of 25 Gt per year between 2025 and 2050.² (In 2023, those emissions amounted to 41 Gt.) The same study reported that meeting that target would require that 33 percent of oil reserves, 49 percent of natural gas reserves, and 82 percent of coal reserves worldwide remain unused through 2050. In the United States, about 6 percent of oil reserves, 4 percent of gas reserves, and 92 percent of coal reserves would remain unused.

Estimated Losses From Stranded Assets

The Congressional Budget Office estimates that the immediate, unanticipated global adoption and enforcement of a 2°C target in 2024 would cause U.S. entities to lose private assets with a

1. Proven reserves are those that could probably be recovered using existing technology under current economic conditions. CBO's estimate is based on reserves reported in International Energy Agency, *World Energy Outlook 2024* (October 2024), Table B.3, www.iea.org/reports/world-energy-outlook-2024; and emissions factors reported in Richard Heede and Naomi Oreskes, "Potential Emissions of CO₂ and Methane From Proved Reserves of Fossil Fuels: An Alternative Analysis," *Global Environmental Change*, vol. 36 (January 2016), pp. 12–20, <https://doi.org/10.1016/j.gloenvcha.2015.10.005>.

2. Christophe McGlade and Paul Ekins, "The Geographical Distribution of Fossil Fuels Unused When Limiting Global Warming to 2°C," *Nature*, vol. 517, no. 7533 (January 8, 2015), pp. 187–190, <https://doi.org/10.1038/nature14016>. The estimates in this paragraph reflect adjustments to the published values to include emissions through 2024.

Continued

like that in SFHAs sell at a discount relative to comparable properties with no flood risk.

Those discounts vary geographically, and they depend on available information about properties and the characteristics of buyers in the market. Some states require home sellers to disclose information about flood risks. Those requirements appear to prompt discounts for homes outside of SFHAs, where buyers might otherwise be unaware of flood risks. Even within flood zones, where properties are already discounted, the requirements seem to have significant price effects. Homes in SFHAs with flood disclosure requirements are discounted by 5 percent to 10 percent, about twice as much as homes in

SFHAs without disclosure requirements.³⁸ The provision of flood hazard information on real estate websites has led to lower prices as well.³⁹

Risks to Property Prices From Wildfires. Home prices tend to reflect wildfire risks when risk disclosures are legally mandated or when the local area has recently experienced a wildfire. Only California and Oregon have state mandates requiring home sellers to disclose wildfire

38. See "Flood Risk and Home Prices" in the appendix.

39. Daryl Fairweather and others, "The Impact of Climate Risk Disclosure on Housing Search and Buying Dynamics: Evidence From a Nationwide Field Experiment With Redfin" (paper presented at the National Bureau of Economic Research Summer Institute, July 24, 2023), <https://tinyurl.com/bfc99v8n>.

Box 3-1.

Continued

Transition Risks

present value of \$430 billion to \$5.5 trillion.³ (That present value reflects the current dollar value of all projected losses.) Those losses would amount to about 0.3 percent to 3.3 percent of the value of net assets held by U.S. households and nonprofit organizations.⁴ One study estimated that in the United States, the wealthiest 10 percent of households would bear about 82 percent of the losses from stranded assets.⁵ Direct losses to the federal government would be relatively small, resulting from lower receipts from the lease of federal lands for oil and gas exploration and a loss in value of the Strategic Petroleum Reserve. Global losses would be about four times as large as U.S. losses, with a present value of \$1.9 trillion to \$24 trillion.

The wider range of estimates of the value of stranded assets in the research literature reflects uncertainty about what assets

to value, how to measure their value, and how their value will change in the future—both with and without climate policies. For example, some estimates of stranded assets account for the retail value of all lost fuel production, including the costs of extraction, refining, and transportation; others reflect the potential profits from selling all known reserves rather than the profits from fuel that could be extracted and sold in a fixed period of time.

Valuations of potentially stranded assets are also sensitive to assumptions about future technological developments and economic growth. Projections of relatively rapid growth of fossil fuel production and high fuel prices in the absence of climate policies yield larger estimated losses. Future prices for alternative fuels are important, too: To the extent that alternative energy sources become cheaper than fossil fuels, those fossil fuels and related assets will be less valuable, and the potential losses from stranded assets will be smaller. Estimates of stranded assets also depend on the time frame over which market participants evaluate future profits and the discount rates they use to calculate the present value of those profits.

The value of stranded assets will further depend on when and how investors learn about, evaluate, and respond to emission-reduction policies. Learning about policies well in advance of their implementation would give investors more time to adjust, so revaluations of assets would be more gradual. To the extent that investors already see such policies as a possibility, asset values will already account for some of the risk of losses, and future losses will be correspondingly smaller.

3. The ranges of estimates provided here are based primarily on estimates published in Gregor Semieniuk and others, “Stranded Fossil Fuel Assets Translate to Major Losses for Investors in Advanced Economies,” *Nature Climate Change*, vol. 12, no. 6 (June 2022), pp. 532–538, <https://doi.org/10.1038/s41558-022-01356-y>; and T. A. Hansen, “Stranded Assets and Reduced Profits: Analyzing the Economic Underpinnings of the Fossil Fuel Industry’s Resistance to Climate Stabilization,” *Renewable and Sustainable Energy Reviews*, vol. 158 (April 2022), article 112144, <https://doi.org/10.1016/j.rser.2022.112144>.
4. Board of Governors of the Federal Reserve System, “Balance Sheet of Households and Nonprofit Organizations, 1952–2023” (accessed May 13, 2024), <https://tinyurl.com/yckcyure>.
5. Gregor Semieniuk and others, “Stranded Fossil Fuel Assets Translate to Major Losses for Investors in Advanced Economies,” *Nature Climate Change*, vol. 12, no. 6 (June 2022), pp. 532–538, <https://doi.org/10.1038/s41558-022-01356-y>.

risks. (There are no federally mapped high-risk zones for wildfires, as there are for floods.) In California, wildfire risk disclosures reduce the price of homes by approximately 4 percent.⁴⁰ Additionally, homes in areas that have recently experienced wildfires sell for less than comparable homes in unaffected areas.⁴¹ That effect dimin-

ishes after a few years, even if the underlying risk remains high.⁴² Fire risk is covered by a standard homeowner’s insurance policy, though, so higher premiums might already be affecting home prices in high-hazard areas.

40. Lala Ma and others, “Risk Disclosure and Home Prices: Evidence From California Wildfire Hazard Zones,” *Land Economics*, vol. 100, no. 1 (February 2024), pp. 6–21, <https://doi.org/10.3368/le.100.1.102122-0087R>.

41. Miyuki Hino and Christopher B. Field, “Fire Frequency and Vulnerability in California,” *PLOS Climate*, vol. 2, no. 2 (February 2023), article e0000087, <https://doi.org/10.1371/journal.pclm.0000087>; and Julie Mueller, John Loomis, and Armando González-Cabán, “Do Repeated Wildfires Change

Homebuyers’ Demand for Homes in High-Risk Areas? A Hedonic Analysis of the Short and Long-Term Effects of Repeated Wildfires on House Prices in Southern California,” *Journal of Real Estate Finance and Economics*, vol. 38, no. 2 (February 2009), pp. 155–172, <https://doi.org/10.1007/s11146-007-9083-1>.

42. Shawn J. McCoy and Randall P. Walsh, “Wildfire Risk, Salience, and Housing Demand,” *Journal of Environmental Economics and Management*, vol. 91 (September 2018), pp. 203–228, <https://doi.org/10.1016/j.jeem.2018.07.005>.

Mortgage Lenders

Mortgage lenders also face some risk from climate damage. When a home with a mortgage is damaged, the remaining loan balance sometimes exceeds the property's value (particularly when the damage was not covered by insurance), which gives the borrower an incentive to default on the mortgage. For instance, homeowners affected by Hurricane Harvey were more likely to default on their mortgages if they did not have flood insurance.⁴³

In addition, mortgage lenders face the risk that property prices will fall because of changing expectations about future damage. If many homes in an area were damaged by flooding from sea-level rise, those losses would probably affect prices for homes that are not yet inundated but face similar threats. If widespread devaluations were large enough, they would affect mortgage lenders enough that the financial difficulties would spread through the financial system. That risk would partly depend on whether the mortgages held by firms were concentrated in affected areas.

Only about a quarter of residential mortgages are held by banks, but those that do hold mortgages have begun to reduce their risk exposure by charging higher interest rates and demanding lower loan-to-value ratios for mortgages on properties that are more vulnerable to flooding or wildfires.⁴⁴ The higher interest rates provide the banks with greater returns to cover any unanticipated losses on mortgages that they hold on their balance sheets. Those higher rates also make it difficult for lower-income borrowers to take out risky mortgages. Requiring lower loan-to-value ratios reduces the risk of default by making it less likely that the value of a damaged home will fall short of the amount still owed on the mortgage.

Risk Transferred to Fannie Mae and Freddie Mac.

Banks shift some of the risk of default or prepayment to the federal government by selling mortgages to Fannie Mae and Freddie Mac—government-sponsored

enterprises that aim to ensure a stable supply of financing for residential mortgages. Both agencies buy mortgages and sell mortgage-backed securities to investors. Those agencies also sell guarantees for the mortgage-backed securities. The guarantees mean that, in the event of borrower default and foreclosure, the agencies must repay the holders of the mortgage-backed securities. Some evidence suggests that lenders are strategically selling mortgages with higher climate risk to Fannie Mae and Freddie Mac.⁴⁵

Local mortgage lenders appear to securitize risk-exposed mortgages more than larger, more diversified lenders, thereby shifting more of their risk to the federal government.⁴⁶ Those local lenders probably have better information about local risks than national lenders and therefore know better which mortgages to securitize to reduce their risk. Larger lenders can more easily diversify their portfolios nationally, reducing their need to securitize mortgages. Some lenders also appear to be shifting risk to other parts of the private sector by using a different kind of mortgage-backed security that is not guaranteed by the federal government, typically for mortgages that are larger than those eligible for securitization by Fannie Mae and Freddie Mac.⁴⁷

Risk Held by Investors in Mortgage-Backed

Securities. Investors in mortgage-backed securities—financial securities whose payments of interest and principal are backed by the payments from a pool of mortgages—also face a growing risk that the value of their investment could fall if homeowners prepay their mortgage after a natural disaster. That would occur if homeowners used their insurance payments to prepay their mortgage and sold or abandoned their property instead of rebuilding it after it was damaged. When homeowners pay off their mortgages early, investors in mortgage-backed securities have to reinvest the money at prevailing interest rates. If those rates are lower than

43. Carolyn Kousky, Mark Palim, and Ying Pan, "Flood Damage and Mortgage Credit Risk: A Case Study of Hurricane Harvey," *Journal of Housing Research*, vol. 29, sup. 1 (November 2020), pp. S86–S120, <https://doi.org/10.1080/10527001.2020.1840131>.

44. Board of Governors of the Federal Reserve System, "Financial Accounts of the United States Release Tables: Mortgage Debt Outstanding, Millions of Dollars; End of Period," FRED database (Federal Reserve Bank of St. Louis, accessed December 8, 2024), <https://tinyurl.com/fykst5cz>. See also "Mortgage Lenders' Climate Risk Management: Stricter Lending Standards" in the appendix.

45. See "Mortgage Lenders' Climate Risk Management: Securitization" in the appendix.

46. Jesse M. Keenan and Jacob T. Bradt, "Underwaterwriting: From Theory to Empiricism in Regional Mortgage Markets in the U.S.," *Climatic Change*, vol. 162 (June 2020), pp. 2043–2067, <https://doi.org/10.1007/s10584-020-02734-1>.

47. Matthew E. Kahn, Amine Ouazad, and Erkan Yönder, *Adaptation Using Financial Markets: Climate Risk Diversification Through Securitization*, Working Paper 32244 (National Bureau of Economic Research, March 2024), www.nber.org/papers/w32244h.

the rate received on the previous mortgage, it reduces the interest income investors receive. Increases in prepayments have frequently occurred after large hurricanes.⁴⁸

Insurance Providers

Insurers bear the risk of losses when natural disasters damage properties covered by their policies. In response to large losses from recent natural disasters, insurers have begun to raise premiums, especially in disaster-prone areas. Increases in premiums shift insurers' risks back to property owners, mortgage lenders, and governments.

In California and some other states, however, regulators have limited insurers' ability to raise premiums. Those regulatory limits on premiums make property insurance more affordable, but they have also led some insurers to restrict the availability of property insurance in high-risk areas or even pull out of markets entirely.⁴⁹ Subsidies are another method governments use to make insurance more affordable, but like limits on premium increases, they incentivize development in risky areas by dampening the risk signal of higher premiums.

Effects of Government Subsidies. Homeowners obtain flood insurance primarily through the National Flood Insurance Program (NFIP). The program provides subsidies for some policies grandfathered into an old pricing model that cause premiums to be lower than the risk-based costs. The NFIP is in the process of phasing out those subsidies, but limits to annual increases in premiums mean that it is only slowly transitioning away from providing federal subsidies for that coverage. (Because of those limits on premium increases, it will take until 2037 for 95 percent of current policies to have premiums that fully account for risk.)⁵⁰ The remaining subsidies make insurance more affordable for property owners who are

required to have coverage, but they also prevent premiums from signaling risk.

In some states where insurance for other natural hazards has become more difficult to obtain, including California, Florida, and Louisiana, state governments have created programs to provide insurance, which in some cases is subsidized. The subsidies allow premiums to be lower than the actual risk would warrant. In the case of a major disaster, the state-run plans in Florida and Louisiana can impose surcharges on all policies in the state to shift some of the costs from policyholders in high-risk areas to those in low-risk areas.

Effects of Regulations. Some state-level regulations on premium increases have prompted insurers to leave high-risk markets where they cannot set risk-based premiums. For example, several large national insurers recently stopped writing new homeowner's policies in California, where regulations prevent insurers from using climate change models to set rates. Evidence from one study indicates that instead of pulling out of those markets, multistate insurance companies may recoup their losses by charging higher rates in states with less regulation.⁵¹ Like government subsidies, those effective subsidies for high-risk properties reduce the ability of premiums to communicate information about risk to property owners.

The Federal Government

The federal government pays for some damage to property through disaster assistance programs and through subsidies to the NFIP. Since 2005, federal spending for disaster relief has averaged tens of billions of dollars per year.⁵² Demand for disaster relief will likely grow as damage from flooding and wildfires increases. And although recent reforms have eliminated most of the subsidies provided by the NFIP, some properties (such as those built before the first flood insurance maps were created) still receive subsidized policies.

48. Carolyn Kousky, Mark Palim, and Ying Pan, "Flood Damage and Mortgage Credit Risk: A Case Study of Hurricane Harvey," *Journal of Housing Research*, vol. 29, sup. 1 (November 2020), pp. S86–S120, <https://doi.org/10.1080/10527001.2020.1840131>; and Justin Gallagher and Daniel Hartley, "Household Finance After a Natural Disaster: The Case of Hurricane Katrina," *American Economic Journal: Economic Policy*, vol. 9, no. 3 (August 2017), pp. 199–228, <https://doi.org/10.1257/pol.20140273>.

49. For a discussion of climate change risks and insurance, see Congressional Budget Office, *Climate Change, Disaster Risk, and Homeowner's Insurance* (August 2024), www.cbo.gov/publication/59918.

50. Government Accountability Office, *Flood Insurance: FEMA's New Rate-Setting Methodology Improves Actuarial Soundness but*

Highlights Need for Broader Program Reform, GAO-23-105977 (July 2023), www.gao.gov/assets/830/828044.pdf.

51. Sangmin S. Oh, Ishita Sen, and Ana-Maria Tenekedjieva, *Pricing of Climate Risk Insurance: Regulation and Cross-Subsidies*, Finance and Economics Discussion Series Paper 2022-064 (Board of Governors of the Federal Reserve System, June 2022), <https://doi.org/10.17016/feds.2022.064>.

52. Congressional Budget Office, *Federal Spending for Flood Adaptations* (September 2024), www.cbo.gov/publication/59971.

In addition, when homeowners default on or prepay their mortgages, some of the losses fall on federal housing programs. Federal mortgage programs do not charge higher rates to guarantee mortgages in areas prone to natural disasters, even though mortgages in those areas have higher rates of default.⁵³ CBO estimates that the annual subsidy cost of flood damage to homes with mortgages guaranteed directly by the federal government will rise by 44 percent by 2053. Under the central estimate of climate conditions, that cost would be about

\$400 million.⁵⁴ To the extent that damage from disasters or increases in insurance premiums drive down the prices of undamaged homes, that cost will be higher. It will also be higher if future flood damage increases prepayment rates, because prepayments decrease income to federal housing programs.

53. Pedro Gete, Athena Tsouderou, and Susan M. Wachter, "Climate Risk in Mortgage Markets: Evidence From Hurricanes Harvey and Irma," *Real Estate Economics*, vol. 52, no. 3 (May 2024), pp. 660–686, <https://doi.org/10.1111/1540-6229.12477>.

54. Evan Herrnsstadt and Byoung Hark Yoo, *The Effects of Flood Damage on the Subsidy Cost of Federally Backed Mortgages*, Working Paper 2024-04 (Congressional Budget Office, July 2024), www.cbo.gov/publication/60167. The subsidy cost is the discounted present value of all future cash flows associated with a loan or loan guarantee. A present value is a single number that expresses the flow of current and future payments in terms of an equivalent lump sum received or paid at a specific time.

Chapter 4: Other Consequences of Climate Change

Climate change will affect the United States in many ways that are not directly captured by changes in gross domestic product but will still have implications for the economy, the federal budget, and society. Those effects include threats to the health of people, plants, and animals and changes in societal factors such as immigration and national security. The magnitude of the effects will depend on the extent and pace of climate change and the extent of adaptation to climate change. Across the United States, the effects of increasingly severe natural disasters and hotter temperatures will be disproportionately borne by low-income and minority households and by people who live in the Southeast.

Human Health

The effects of climate change on human health will depend on both the extent of climate change and the extent of adaptation to it. Extreme temperatures are associated with a variety of negative health effects. Climate change will increase the frequency of extremely hot days, which will raise mortality rates, and decrease the frequency of extremely cold days, which will lower them. In addition to the direct effects of temperature on health, climate change will indirectly worsen health outcomes by increasing pollution and expanding the range of organisms that transmit disease. Because older people are more vulnerable to extreme temperatures, pollution, and disease, the continued aging of the population will amplify the effects on mortality regardless of how much temperatures rise. The impacts on health and mortality rates will affect costs for health care programs and Social Security.

Given warming of 4 degrees Celsius by 2100 with no further adaptation, U.S. mortality rates would be 1.5 percent to 2.0 percent higher than they would be in the absence of continued climate change; further aging of the population would make the effect even larger.¹

(In 2022, a 1 percent increase in mortality would have amounted to about 35,000 deaths, and an increase of 1.5 percent to 2.0 percent would have caused about as many deaths as kidney disease.)² But adaptation will mitigate or perhaps even reverse that rise in deaths. Accounting for the effects of adaptation, estimates of the change in mortality rates with the same amount of warming range from a 0.5 percent increase to a 0.5 percent decrease. For warming of about 2°C and no further adaptation, estimates of the effect on mortality rates run from an increase of 0.5 percent to a decrease of about 1.0 percent. With sufficient adaptation, the reduction in deaths due to cold weather is expected to outweigh the increase in deaths due to hot weather, resulting in a 0.5 percent to 1.5 percent reduction in mortality rates.

Those estimates reflect changes in the number of deaths due to extreme temperatures. The studies that reported them were not designed to capture the mortality effects of certain longer-term effects of climate change, such as increased wildfire pollution or expanded ranges of disease-carrying organisms. For that reason, the estimates probably underrepresent the full effects of climate change on mortality.

Hot Weather

Climate change will increase rates of injury, illness, and death due to extreme heat. Heat stress, which occurs when the body is not able to adequately cool itself, negatively affects many of the body's functions. It worsens health conditions such as heart disease, diabetes, and respiratory diseases and sometimes causes kidney failure. Heat stress is most dangerous for adults over 65 and young children—especially infants. Because of those effects, hot temperatures are associated with increases in

1. Some of the studies used to estimate the effect of mortality in this report did not account for future changes in the average age of the population. See “National Estimates of Temperature-Related Mortality” in the appendix.

2. Kenneth D. Kochanek and others, *Mortality in the United States, 2022*, Data Brief 492 (National Center for Health Statistics, March 2024), www.cdc.gov/nchs/products/databriefs/db492.htm.

mortality rates and emergency department visits.³ (One study estimated that in California, emergency department visits would increase by 0.5 percent by 2050 if global warming reached 2°C.)⁴ Beyond physical illness, rates of mental illness are expected to rise by about 1 percent for each degree of warming, and some research suggests that suicides will rise by 3 percent if warming reaches 4°C and by 1.5 percent if warming reaches 2°C.⁵ (In 2022, suicide represented about 1.5 percent of overall U.S. mortality, claiming about 50,000 lives.)⁶

Americans are likely to adapt to hotter climates. Between the beginning and the end of the 20th century, adaptations to extreme heat reduced its effect on mortality by roughly 90 percent.⁷ Increases in mortality on hot days are much smaller in hot regions, and increases in mortality on cold days are much smaller in cold regions. As temperatures increase, adaptations that are now used in the hottest parts of the country will be more widely adopted in colder places. Those adaptations include more use of air conditioning or other cooling technologies, as well as physical acclimatization and behavioral changes (for example, shifting outdoor activity to mornings and evenings).

3. Corey White, "The Dynamic Relationship Between Temperature and Morbidity," *Journal of the Association of Environmental and Resource Economists*, vol. 4, no. 4 (December 2017), pp. 1155–1198, <https://doi.org/10.1086/692098>.
4. Carlos F. Gould and others, *Temperature Extremes Impact Mortality and Morbidity Differently*, Working Paper 32195 (National Bureau of Economic Research, March 2024), www.nber.org/papers/w32195.
5. Marshall Burke and others, "Higher Temperatures Increase Suicide Rates in the United States and Mexico," *Nature Climate Change*, vol. 8, no. 8 (August 2018), pp. 723–729, <https://doi.org/10.1038/s41558-018-0222-x>.
6. Centers for Disease Control and Prevention, "Web-Based Injury Statistics Query and Reporting System" (accessed July 30, 2024), www.cdc.gov/injury/wisqars/index.html; and Kenneth D. Kochanek and others, *Mortality in the United States, 2022*, Data Brief 492 (National Center for Health Statistics, March 2024), www.cdc.gov/nchs/products/databriefs/db492.htm.
7. Alan Barreca and others, "Convergence in Adaptation to Climate Change: Evidence From High Temperatures and Mortality, 1900–2004," *American Economic Review*, vol. 105, no. 5 (May 2015), pp. 247–251, <https://doi.org/10.1257/aer.p20151028>. See also Alan Barreca and others, "Adapting to Climate Change: The Remarkable Decline in the U.S. Temperature–Mortality Relationship Over the Twentieth Century," *Journal of Political Economy*, vol. 124, no. 1 (February 2016), pp. 105–159, <https://doi.org/10.1086/684582>.

Adaptations will entail trade-offs. One study estimated that if global warming reached 4°C, spending on adaptive measures and losses due to behavioral trade-offs in the United States would total about 1 percent of GDP annually by 2100.⁸ (For reference, 1 percent of GDP in 2023 was \$274 billion.) For 2°C of warming, those costs were estimated to equal about 0.2 percent of GDP annually by 2100. The costs of adaptation—and the increases in mortality with less adaptation—will be greater if the effects of unprecedented temperature extremes on mortality are stronger than expected.

Although adaptations strongly reduce the effect of heat on mortality, they will not benefit everyone equally. People who work outdoors will be particularly vulnerable to rising temperatures; so will people who remain without access to air conditioning. (In 2020, about one in ten U.S. households did not have air conditioning; among households with income less than \$20,000, that share was roughly one in five.)⁹ The moderating effects of adaptations that depend on electricity will also be diminished if natural disasters disrupt the electrical grid.

Cold Weather

Climate change will reduce rates of injury, illness, and death due to cold weather. Extremely cold weather causes death through hypothermia, but for the most part, cold weather influences mortality by worsening preexisting medical conditions such as cardiovascular and respiratory diseases. Death rates from heart attacks increase when temperatures are low, likely because cold constricts blood vessels and hinders circulation.

Pollution

Beyond the direct effect of temperature on mortality, climate change will affect health by increasing air pollution. Hot, sunny conditions speed up the chemical reactions that produce ground-level ozone, the main component of smog. (Smog's short-term effect on mortality is accounted for in the estimates above, but its long-term effects on the lungs, heart, blood vessels, and immune system are not.) Trees remove particulate matter from the

8. Tamma Carleton and others, "Valuing the Global Mortality Consequences of Climate Change Accounting for Adaptation Costs and Benefits," *Quarterly Journal of Economics*, vol. 137, no. 4 (November 2022), pp. 2037–2105, <https://doi.org/10.1093/qje/qjac020>.
9. Department of Energy, "Residential Energy Consumption Survey: 2020 RECS Survey Data" (March 2023), Table HC7.5, <https://tinyurl.com/ytz23cbn>.

atmosphere, so in places that lose trees because of climate change, air pollution will worsen. In addition, hotter, drier weather creates conditions that increase the risk of wildfires, including larger and more severe fires that produce more particulate matter.

One study estimated that the impact of wildfire pollution on mortality will be as large as the impact of heat stress without the mitigating effects of adaptations to hot weather.¹⁰ Currently, estimates of deaths from exposure to wildfire smoke in the United States range widely, from 6,000 to 30,000 per year, accounting for 0.2 percent to 0.9 percent of annual mortality.¹¹ According to another study, by 2050, annual deaths due to wildfire smoke will increase by 51 percent to 76 percent.¹²

In general, the effects of climate change on air quality are expected to vary widely across regions and groups. Impacts on air quality will be particularly detrimental in heat- and drought-prone states, and adults over 65 and children will face the greatest health risks. Efforts to reduce emissions in the electric power and transportation sectors would offset some increases in pollution from climate change.

Vector-Borne Diseases

Climate change will further affect health by expanding the geographic range of disease-carrying organisms such as mosquitoes, ticks, and rats, which is likely to increase U.S. cases of Dengue fever, West Nile virus, Lyme disease, and other diseases.¹³ The magnitude of those increases will vary depending on the type of disease and the extent of climate change. For example, with global warming of 3°C or more by 2100, cases of West Nile virus are expected to roughly double; with warming between 2°C and 3°C, the increase is expected to be

about half as large, amounting to about 1,000 additional cases per year.¹⁴

Biodiversity

The biological diversity of life (sometimes referred to as biodiversity) provides a wide range of benefits to the United States.¹⁵ Healthy, diverse ecosystems support all sources of food and nature-based materials by providing water, nutrients, and shelter for animals, plants, marine life, and other organisms. By altering the climatic conditions upon which plants and animals depend, climate change threatens to degrade ecosystems and increase extinction risk. Because the pace of current and projected climate change is unusually rapid in geologic terms, warming temperatures and related impacts are likely to alter environments more quickly than many organisms can adapt or evolve in response. Such changes would present risks to agriculture, to health, and to protections from natural disasters, all of which would affect the economy and the federal budget.

Effects on Ecosystems

Climate change is likely to have both beneficial and harmful effects on ecosystems, but the harmful effects are expected to dominate. For example, rising atmospheric concentrations of carbon dioxide will stimulate more photosynthesis in plants, and growing seasons will lengthen in some regions. Nevertheless, intensifying heat waves, wildfires, and droughts will progressively degrade ecosystems in other areas. Oceans are at particular risk: The absorption of carbon dioxide will make ocean water more acidic, with generally harmful effects on marine organisms, especially corals and shellfish.

Species typically survive only within a certain range of environmental conditions. Changes in climate are expected to increase extinction rates in part by taking away territory in which species experience those conditions. Expanded agriculture, deforestation, and

10. Marshall Burke and others, “The Changing Risk and Burden of Wildfire in the United States,” *Proceedings of the National Academy of Sciences of the United States of America*, vol. 118, no. 2 (January 12, 2021), article e2011048118, <https://doi.org/10.1073/pnas.2011048118>.

11. See “Wildfires and Mortality” in the appendix.

12. Minghao Qiu and others, *Mortality Burden From Wildfire Smoke Under Climate Change*, Working Paper 32307 (National Bureau of Economic Research, April 2024), www.nber.org/papers/w32307.

13. Mary H. Hayden and others, “Human Health,” in Allison R. Crimmins and others, eds., *Fifth National Climate Assessment* (U.S. Global Change Research Program, November 2023), Chapter 15, <https://nca2023.globalchange.gov/chapter/15>.

14. Environmental Protection Agency, *Multi-Model Framework for Quantitative Sectoral Impacts Analysis: A Technical Report for the Fourth National Climate Assessment*, EPA 430-R-17-001 (May 2017), <https://tinyurl.com/m4jjd4k3>.

15. Some researchers are working to augment traditional measures of national wealth (physical capital and human capital) to include what they call natural capital—natural resources such as fossil fuels, metals, and minerals; forests, cropland, and pastureland; wetlands and marine fisheries; and perhaps even living organisms. See, for example, World Bank, *The Changing Wealth of Nations, 2021: Managing Assets for the Future* (October 2021), <http://hdl.handle.net/10986/36400>.

urbanization will continue to put pressure on ecosystems as well.¹⁶ Such human activities will continue to contribute to losses of biodiversity and other changes to ecosystems, but by 2070, climate change is likely to become the dominant cause of biodiversity loss.¹⁷

Organisms have some ability to adapt to changing environmental conditions by migrating, changing their behavior, or evolving. As noted above, however, climate change is projected to unfold much more rapidly than it has in the past. For instance, by the end of the century, North American climate zones will shift several hundreds of miles north from where they are today.¹⁸ That rapid pace of change is likely to require a speed of migration that is unrealistic for amphibians, reptiles, and plants.¹⁹ Geographic barriers and human land development will also limit animals' ability to migrate. For example, species seeking cooler climates in alpine areas such as the Rocky Mountains will have little additional area at higher elevations that is suitable for their survival.

According to the Intergovernmental Panel on Climate Change, with 4°C of warming, 13 percent of species would lose more than 80 percent of their climatically suitable habitat, which would put them at high risk of extinction. With 2°C of warming, 10 percent would face such a risk. However, because climate change could have a variety of uncertain effects on the survival of animals and plants, estimates of its effects on extinction rates have a wide range. Under the central estimate of

warming, for example, there is a 5 percent chance that the share of species at very high risk of extinction would be 37 percent and a 5 percent chance that it would be as small as 2 percent.²⁰ Limited data on the species that are most vulnerable to the effects of climate change—including invertebrates, amphibians, and flowering plants—suggest that the true number of species at risk of extinction is larger.²¹

Some species that are not threatened with global extinction will nevertheless be lost to local ecosystems. Some individual ecosystems will therefore be seriously affected by the loss of key species that serve roles such as predator, prey, decomposer, pollinator, or seed disseminator. Notwithstanding the ability of species to adapt, local ecosystems are expected to lose 38 percent of their terrestrial vertebrate species with warming of about 4°C and 20 percent of their terrestrial vertebrate species if warming is limited to roughly 2°C.²² Even if species had an unlimited ability to migrate, the ecosystems created by climate change are likely to have less biodiversity than ecosystems today.²³

Economic Risks of Biodiversity Loss

Losses to global biodiversity and the resulting degradation of ecosystems present several economic risks—including risks to agriculture, to health, and to protections from natural disasters—that would also affect the federal budget through changes in tax revenues and

16. Over the past 500 years, such activities have caused extinctions to occur at more than 100 times the rate experienced in the preceding 10 million years. See Gerardo Ceballos and others, "Accelerated Modern Human-Induced Species Losses: Entering the Sixth Mass Extinction," *Science Advances*, vol. 1, no. 5 (June 19, 2015), <https://doi.org/10.1126/sciadv.1400253>.

17. Tim Newbold, "Future Effects of Climate and Land-Use Change on Terrestrial Vertebrate Community Diversity Under Different Scenarios," *Proceedings of the Royal Society B: Biological Sciences*, vol. 285, no. 1881 (June 27, 2018), <https://doi.org/10.1098/rspb.2018.0792>.

18. Matthew C. Fitzpatrick and Robert R. Dunn, "Contemporary Climatic Analogs for 540 North American Urban Areas in the Late 21st Century," *Nature Communications*, vol. 10 (February 2019), article 614, <https://doi.org/10.1038/s41467-019-08540-3>.

19. Migration at the required pace is feasible only for some mammals, birds, fish, and insects. See Alex L. Pigot and others, "Abrupt Expansion of Climate Change Risks for Species Globally," *Nature Ecology and Evolution*, vol. 7, no. 7 (July 2023), pp. 1060–1071, <https://doi.org/10.1038/s41559-023-02070-4>.

20. Camille Parmesan and others, "Terrestrial and Freshwater Ecosystems and Their Services," in Hans-Otto Pörtner and others, eds., *Climate Change 2022: Impacts, Adaptation, and Vulnerability, Working Group II Contribution to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* (Cambridge University Press, 2022), pp. 197–378, <https://doi.org/10.1017/9781009325844>.

21. Partha Dasgupta, *The Economics of Biodiversity: The Dasgupta Review* (HM Treasury, February 2021), <https://tinyurl.com/3xrsanuh>.

22. Tim Newbold, "Future Effects of Climate and Land-Use Change on Terrestrial Vertebrate Community Diversity Under Different Scenarios," *Proceedings of the Royal Society B: Biological Sciences*, vol. 285, no. 1881 (June 27, 2018), <https://doi.org/10.1098/rspb.2018.0792>.

23. Pamela D. McElwee and others, "Ecosystems, Ecosystem Services, and Biodiversity," in Allison R. Crimmins and others, eds., *Fifth National Climate Assessment* (U.S. Global Change Research Program, November 2023) <https://doi.org/10.7930/NCA5.2023.CH8>.

spending (for example, for crop insurance, health care, and disaster assistance programs).²⁴

- Risks to agriculture. At roughly 4°C of warming, over 30 percent of the land used to grow crops and raise livestock worldwide is projected to become climatically unsuitable for agriculture, though other areas are expected to become more suitable.²⁵ Reductions in global crop yields are expected to pose a risk to agricultural imports and put inflationary pressure on food prices by making some types of foods scarcer. Crops such as cacao, coffee, and many types of fruits and vegetables depend on insects to pollinate them, so losses of pollinators due to climate change pose a risk to agricultural output.²⁶
- Risks to health. As many as half of prescription drugs are based on naturally occurring molecules, and 70 percent of cancer drugs are created from plants or other organisms. Most of those compounds come from tropical locations that are at higher risk from climate change.²⁷
- Risks to protections from natural disasters. Coral reefs currently function as breakwaters for many U.S. coastlines, preventing over \$1 billion in flood damage to U.S. properties each year.²⁸ Because corals are particularly sensitive to the temperature and acidity of

their environments, even 2°C of warming will lead to the near-total loss of U.S. reef coverage.²⁹

Immigration

In many parts of the world, climate change will increase agricultural losses and damage due to natural disasters. Both effects are likely to cause more people to immigrate to the United States than would do so in the absence of further climate change.³⁰ Such an increase would probably boost output and revenues as well as federal spending.³¹ However, the influence of climate change on U.S. immigration will probably be smaller than the influence of the economic, political, and social conditions that also prompt migration, such as poverty, conflict, and family reunification.³² And even if climate change increases people's desire to immigrate to the United States, their ability to do so will be limited by geography, resources, and U.S. immigration policy.

Because patterns of immigration depend on a complex set of social and economic factors, estimates of climate-related immigration are uncertain. Natural disasters and droughts have increased immigration in some cases but have reduced it in others, which indicates that the influence of climate change on immigration

24. For an examination of investors' responses to the risks posed by biodiversity loss, see Stefano Giglio and others, *The Economics of Biodiversity Loss*, Working Paper 32678 (National Bureau of Economic Research, July 2024), www.nber.org/papers/w32678.

25. Rachel Bezner Kerr and others, "Food, Fibre, and Other Ecosystem Products," in Hans-Otto Pörtner and others, eds., *Climate Change 2022: Impacts, Adaptation, and Vulnerability, Working Group II Contribution to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* (Cambridge University Press, 2022), pp. 713–906, <https://doi.org/10.1017/9781009325844>.

26. See, for example, Joseph Millard and others, "Key Tropical Crops at Risk From Pollinator Loss Due to Climate Change and Land Use," *Science Advances*, vol. 9, no. 41 (October 12, 2023), <https://doi.org/10.1126/sciadv.adh0756>.

27. World Economic Forum, *Nature Risk Rising: Why the Crisis Engulfing Nature Matters for Business and the Economy* (January 2020), <https://tinyurl.com/56m66xp2>.

28. Curt D. Storlazzi and others, *Rigorously Valuing the Role of U.S. Coral Reefs in Coastal Hazard Risk Reduction*, Open-File Report 2019-1027 (U.S. Geological Survey, April 2019), <https://doi.org/10.3133/ofr20191027>.

29. Environmental Protection Agency, *Multi-Model Framework for Quantitative Sectoral Impacts Analysis: A Technical Report for the Fourth National Climate Assessment*, EPA 430-R-17-001 (May 2017), <https://tinyurl.com/3zu3un6y>.

30. CBO's current projections of the U.S. population capture the historical effects of natural disasters and droughts on immigration but not the potential effects of increasingly intense future climate impacts. For those projections, see Congressional Budget Office, *The Demographic Outlook: 2024 to 2054* (January 2024), www.cbo.gov/publication/59697.

31. For a discussion of the economic and budgetary effects of recent increases in immigration, see Congressional Budget Office, *Effects of the Immigration Surge on the Federal Budget and the Economy* (July 2024), www.cbo.gov/publication/60165.

32. Etienne Piguet, "Linking Climate Change, Environmental Degradation, and Migration: An Update After 10 Years," *Wiley Interdisciplinary Reviews: Climate Change*, vol. 13, no. 1 (January/February 2022), article e746, <https://doi.org/10.1002/wcc.746>; Michał Burzyński, Frédéric Docquier, and Hendrik Scheewel, "The Geography of Climate Migration," *Journal of Demographic Economics*, vol. 87, no. 3 (March 2021), pp. 345–381, <https://doi.org/10.1017/dem.2021.6>; and Abrahm Lustgarten, "The Great Climate Migration," *New York Times Magazine* (July 23, 2020), <https://tinyurl.com/yy6sr6jt>.

will be highly dependent on circumstances.³³ Moreover, determining whether people migrate because of climate conditions or because of social and economic factors is challenging—especially because climate change can create or amplify those factors. Historical relationships between climate change and migration can inform estimates of future immigration flows, but short-term responses to weather extremes and natural disasters do not always predict long-term responses, and climate conditions expected by 2100 have no precedent in recorded history.

Climate change is also likely to influence domestic migration within the United States as residents move away from areas at increasing risk of extreme temperatures or natural disasters. Intensified hurricanes have already prompted people to relocate—for example, roughly 4 percent of the population of Puerto Rico moved to the mainland United States in the wake of Hurricane Maria in 2017.³⁴ In the Southwest, declining water resources will likewise increase the pressure to migrate.

Effects of Agricultural Losses

One way in which climate change can affect immigration is that hotter temperatures and droughts can damage agriculture in other countries. Reductions in crop yields due to climate change are expected to be largest in tropical regions that are already quite hot. In the past, the effects of changes in climate on migration have been strongest in the most agriculturally dependent countries, where reductions in crop yields are more likely to cause poverty, food scarcity, and even violence.³⁵ The people in those countries who are most likely to be affected are those in rural areas who rely on agriculture for their subsistence. Some of those people will not be able to afford or access agricultural adaptations such as hardier

alternative crops. However, they are also the least likely to have the resources to migrate.³⁶

Effects of Natural Disasters

Another way in which climate change will affect immigration is through coastal flooding and stronger hurricanes. One study projected that an increase of 2 feet to 3 feet in global sea levels would displace as much as half a percent of the world's population by 2100.³⁷

In the past, hurricanes have caused substantial increases in immigration flows to the United States when they have occurred in countries with large immigrant populations already living here.³⁸ Countries in Latin America and the Caribbean, from which many immigrants have already come to the United States, are expected to face greater damage from hurricanes as sea levels rise and ocean temperatures increase. Nonetheless, many people who are displaced by natural disasters travel relatively short distances and return to their homes relatively quickly because they wish to rebuild.³⁹ It is difficult to predict whether those patterns will continue in the future as disasters intensify.

Effects of Immigration Policies and Migrant Networks

In the United States, some types of immigration are subject to annual caps determined by law, which will limit the number of people who can immigrate for reasons related to climate change. However, other types of immigration are not capped. For example, there is no cap

33. Roman Hoffmann and others, "A Meta-Analysis of Country-Level Studies on Environmental Change and Migration," *Nature Climate Change*, vol. 10, no. 10 (October 2020), pp. 904–912, <https://doi.org/10.1038/s41558-020-0898-6>.

34. Rolando J. Acosta and others, "Quantifying the Dynamics of Migration After Hurricane Maria in Puerto Rico," *Proceedings of the National Academy of Sciences of the United States of America*, vol. 117, no. 51 (December 22, 2020), <https://doi.org/10.1073/pnas.2001671117>.

35. Ruohong Cai and others, "Climate Variability and International Migration: The Importance of the Agricultural Linkage," *Journal of Environmental Economics and Management*, vol. 79 (September 2016), pp. 135–151, <https://doi.org/10.1016/j.jeem.2016.06.005>.

36. Hélène Benveniste, Michael Oppenheimer, and Marc Fleurbaey, "Climate Change Increases Resource-Constrained International Immobility," *Nature Climate Change*, vol. 12, no. 7 (July 2022), pp. 634–641, <https://doi.org/10.1038/s41558-022-01401-w>; and Cristina Cattaneo and Giovanni Peri, "The Migration Response to Increasing Temperatures," *Journal of Development Economics*, vol. 122 (September 2016), pp. 127–146, <https://doi.org/10.1016/j.jdevco.2016.05.004>.

37. Klaus Desmet and others, "Evaluating the Economic Cost of Coastal Flooding," *American Economic Journal: Macroeconomics*, vol. 13, no. 2 (April 2021), pp. 444–486, <https://doi.org/10.1257/mac.20180366>.

38. Parag Mahajan and Dean Yang, "Taken by Storm: Hurricanes, Migrant Networks, and U.S. Immigration," *American Economic Journal: Applied Economics*, vol. 12, no. 2 (April 2020), pp. 250–277, <https://doi.org/10.1257/app.20180438>.

39. Cristina Cattaneo and others, "Human Migration in the Era of Climate Change," *Review of Environmental Economics and Policy*, vol. 13, no. 2 (Summer 2019), pp. 189–206, <https://doi.org/10.1093/reep/rez008>.

on the number of spouses, children, and parents of U.S. citizens who can be admitted for legal, permanent immigration, which means that immigrants are most likely to come from countries with large immigrant populations already present here.

Administrative and judicial decisions also affect the number of immigrants admitted to the United States in certain uncapped categories. For example, people whose home countries have suffered a natural disaster are sometimes granted temporary protected status, which provides protection from deportation. People fleeing violence sparked by drought, heat waves, or disasters may be eligible for asylum, a permanent legal status granted by judges. (The influence of climate change on conflict is discussed below.) Over the next century, the number of immigrants in those groups could increase as the physical impacts of climate change worsen.

National Security

Climate change is expected to affect U.S. military operations by making conflict more likely, by changing elements of defense strategy, and by raising the costs of current operations, all of which are likely to increase the costs of defense in the future.

Increasing Conflict

Climate change is likely to increase geopolitical instability, putting the United States at greater risk of being drawn into a conflict. Although conflict is more likely to be precipitated by factors such as low socioeconomic development, weak governance, and prior unrest, climate change amplifies the effects of those existing risk factors. Extreme weather events such as droughts and natural disasters are expected to make resources scarcer, potentially leading to conflict through economic losses, population movements, or territorial disputes.

Whether the United States will be drawn into such conflicts is a separate question. That is likely to depend on the extent of the conflicts and the countries in which they occur. Evidence suggests that climate change will do more to spur small-scale conflicts than large-scale conflicts such as wars.⁴⁰ The ways in which climate change

affects conflict will be highly dependent on context. For example, a nation's capacity to effectively manage a regional drought and avoid conflict depends on its water conservation and irrigation technology, its agricultural diversity, and its relationships with trading partners. The regions that will be more susceptible to conflict as a result of climate change are those that are more dependent on agriculture and less able to adapt.⁴¹

Predicting the impact of climate change on geopolitical stability is challenging. In one survey, 11 climate and conflict experts estimated that between 3 percent and 20 percent of the risks associated with organized armed conflict within countries over the past century had been influenced by variations in the climate.⁴² However, those past effects will not necessarily predict future political and social reactions to climate change that is unprecedented in recorded history. The same experts estimated that the risks associated with conflict would rise by about 26 percentage points, on average, if global warming reached 4°C and by roughly 13 percentage points if it reached 2°C. Future economic development would reduce those risks.

Changing Elements of Defense Strategy

As environmental conditions change in regions of preexisting U.S. military involvement, the United States will probably have to alter elements of its defense strategy. Further alterations may be needed if climate change creates new areas of military involvement. A key example of that phenomenon is the opening of the Arctic, which was made possible by melting sea ice and has raised the potential for conflict over shipping lanes and opportunities for resource extraction. Addressing new threats in the Arctic region will likely require an increased military presence as well as further investment in specialized technology such as icebreakers.⁴³ Elsewhere in the world, patterns of geopolitical alignment and military strategy

40. Hans-Otto Pörtner and others, "2022: Technical Summary," in Hans-Otto Pörtner and others, eds., *Climate Change 2022: Impacts, Adaptation, and Vulnerability, Working Group II Contribution to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* (Cambridge University Press, 2022), pp. 37–118, <https://doi.org/10.1017/9781009325844.002>.

41. Vally Koubi, "Climate Change and Conflict," *Annual Review of Political Science*, vol. 22 (May 2019), pp. 343–360, <https://doi.org/10.1146/annurev-polisci-050317-070830>.

42. In that study, the risks associated with conflict included increases in the likelihood, frequency, or duration of conflicts and the extent of their harmful consequences (for example, the number of deaths). See Katharine J. Mach and others, "Climate as a Risk Factor for Armed Conflict," *Nature*, vol. 571, no. 7764 (July 11, 2019), pp. 193–197, <https://doi.org/10.1038/s41586-019-1300-6>.

43. Congressional Budget Office, *The Cost of the Coast Guard's Polar Security Cutter* (August 2024), www.cbo.gov/publication/60170.

could shift as a result of state and societal reactions to climate change and national climate policies.

Increasing Costs of Current Operations

Increasing damage to defense installations and changing energy requirements are likely to raise the costs of ongoing defense operations in many places. Several factors have the potential to drive up costs:

- Extreme temperatures, sea-level rise, wildfires, and thawing permafrost are all expected to increase damage to U.S. military installations. For example, more than 1,700 military installations are in coastal areas that are vulnerable to sea-level rise and hurricanes.⁴⁴ One report that assessed the exposure of military installations to climate hazards found that with more climate change, almost all of the sites surveyed would have greater exposure in 2085 than they do today.⁴⁵
- Training and testing activities will be more frequently disrupted by extreme temperatures and natural disasters. Furthermore, as those activities become more likely to spark wildfires, the Department of Defense's spending to address wildfires will probably increase.
- Hotter temperatures pose additional risks to defense operations near the equator. As one example, water service failures present a growing threat to defense installations outside of the United States where the U.S. military's requirements for water compete with local needs.
- Military medical operations in some places will probably be affected by reductions in the supply of clean water as well as by a greater prevalence of heat-related illnesses.

Distributional Effects

The impacts of climate change will be unevenly distributed across the United States. Low-income and minority communities will be more exposed and more vulnerable to those effects; they will also face greater barriers to recovery. Residents of the Southeast are expected to be especially affected, both physically and financially, by increasingly severe natural disasters and hotter temperatures. As a result, the budgetary costs for government

assistance programs that are means-tested (those that provide cash payments or other assistance to people with relatively low income or few assets) will probably increase more than they would if the impacts of climate change were distributed evenly.

Effects by Income and by Race and Ethnicity

Low-income and minority communities are disproportionately exposed to the physical impacts of climate change. Flood damage is expected to be concentrated in low-income communities, especially those in coastal Louisiana, Appalachia, and rural counties of the Northeast and Pacific Northwest.⁴⁶ Low-income and minority groups are more likely to live in high-risk flood zones, perhaps because they lack access to information about the risks, because they cannot afford comparable homes in safer areas, or because of housing discrimination.⁴⁷ Following a natural disaster, lower-income people and Black or Hispanic people are more likely to experience work disruptions, property damage, or the death of a friend or family member.⁴⁸ Low-income families are also less likely to move away from the affected area after a disaster.⁴⁹

Beyond their exposure to natural disasters, low-income households and minority households are also more exposed to heat. In urban areas, temperatures are amplified by densely built infrastructure, which absorbs heat, and human activity, which emits heat. That effect, known as the urban heat island effect, typically raises daytime temperatures in urban areas over those in surrounding rural and suburban areas by 1 to 7 degrees

44. Margaret Tucker and G. James Herrera, *Military Installations and Sea-Level Rise*, Report IF11275, version 3 (Congressional Research Service, July 26, 2019), <https://tinyurl.com/3xttmfk3>.

45. A. O. Pinson and others, *DoD Installation Exposure to Climate Change at Home and Abroad* (Army Corps of Engineers, April 2021), www.acq.osd.mil/eie/eer/cr/cc/resources.html.

46. Congressional Budget Office, *Flood Damage and Federally Backed Mortgages in a Changing Climate* (November 2023), www.cbo.gov/publication/59379; Congressional Budget Office, *Communities at Risk of Flooding* (September 2023), www.cbo.gov/publication/58953; and Oliver E. J. Wing and others, "Inequitable Patterns of U.S. Flood Risk in the Anthropocene," *Nature Climate Change*, vol. 12, no. 2 (February 2022), pp. 156–162, <https://doi.org/10.1038/s41558-021-01265-6>.

47. Laura A. Bakkensen and Lala Ma, "Sorting Over Flood Risk and Implications for Policy Reform," *Journal of Environmental Economics and Management*, vol. 104 (November 2020), article 102362, <https://doi.org/10.1016/j.jeem.2020.102362>.

48. Board of Governors of the Federal Reserve System, *Economic Well-Being of U.S. Households in 2021* (May 2022), <https://tinyurl.com/4wr2z5xd>.

49. Brigitte Roth Tran and Tamara Lynn Sheldon, "Same Storm, Different Disasters: Consumer Credit Access, Income Inequality, and Natural Disaster Recovery" (SSRN, May 28, 2019), <http://dx.doi.org/10.2139/ssrn.3380649>.

Fahrenheit.⁵⁰ The effect is generally strongest in low-income areas, which frequently lack cooling vegetation: On average, low-income areas are 2°F hotter than higher-income areas in the same cities. Across income groups, Black and Hispanic people are more likely than White people to live in areas with intense urban heat island effects.⁵¹ People experiencing homelessness are especially vulnerable to extreme heat. In Phoenix in 2023, for example, nearly half of all deaths attributed to heat occurred among people without housing.⁵²

Adaptations are able to mitigate some of the negative effects of climate change, but low-income communities are less likely to be able to afford them. Some low-income communities exposed to hurricanes and floods, for example, will not have the resources to invest in adaptations such as seawalls and home elevations. The Army Corps of Engineers and the Federal Emergency Management Agency pay for disaster adaptation projects, but those projects are screened on the basis of benefit-cost analyses, which can limit their implementation in communities with low property values.⁵³ Some FEMA grants for projects to mitigate damage from disasters are awarded through a competitive process or require recipients to submit inventories of damage, which can pose administrative challenges for low-income communities.⁵⁴

The vulnerability of low-income and minority communities to physical climate risks will be compounded by a lack of insurance and access to credit, which would help those communities recover financially in the wake of natural

disasters. Households with less income have lower take-up rates of flood insurance, perhaps in part because National Flood Insurance Program plans are overpriced for some low-value properties.⁵⁵ Following natural disasters, people in low-income and minority communities are more likely to experience declines in credit scores, to be delinquent on mortgages and credit cards, and to declare bankruptcy.⁵⁶

Finally, the income and living expenses of low-income households will be disproportionately affected by climate change. Lost earnings due to extreme heat are concentrated in sectors that are exposed to the outdoors and pay wages lower than the national median, such as agriculture, construction, and manufacturing.⁵⁷ Subsequent reductions in labor income are larger in poorer counties.⁵⁸ More broadly, rising costs for food and energy will have an outsized effect on low-income households because basic needs take up a larger share of their budgets. Therefore, low-income households will disproportionately suffer from inflationary pressures on food prices due to reductions in crop yields; they will also disproportionately suffer from higher cooling costs (though they will disproportionately benefit from lower heating costs).

Effects by Region

The physical impacts of climate change are expected to be largest in the Southeast, where increases in already high temperatures and humidity will be especially

50. K. A. Hibbard and others, “Changes in Land Cover and Terrestrial Biogeochemistry,” in D. J. Wuebbles and others, eds., *Climate Science Special Report: Fourth National Climate Assessment*, vol. 1 (U.S. Global Change Research Program, 2017), Chapter 10, <https://doi.org/10.7930/J0416V6X>.

51. Angel Hsu and others, “Disproportionate Exposure to Urban Heat Island Intensity Across Major U.S. Cities,” *Nature Communications*, vol. 12, no. 1 (May 2021), article 2721, <https://doi.org/10.1038/s41467-021-22799-5>.

52. Maricopa County Department of Public Health, *2023 Heat Related Deaths Report* (April 2024), <https://tinyurl.com/mr3drdev>.

53. Congressional Budget Office, *Federal Spending for Flood Adaptations* (September 2024), www.cbo.gov/publication/59971.

54. Benjamin M. Miller and others, *The Cost of Cost-Effectiveness: Expanding Equity in Federal Emergency Management Agency Hazard Mitigation Assistance Grants* (RAND Corporation, Homeland Security Operational Analysis Center, February 2023), www.rand.org/pubs/research_reports/RRA2171-1.html.

55. Carolyn Kousky, “Financing Flood Losses: A Discussion of the National Flood Insurance Program,” *Risk Management and Insurance Review*, vol. 21, no. 1 (Spring 2018), pp. 11–32, <https://doi.org/10.1111/rmir.12090>; and Ajita Atreya, Susana Ferreira, and Erwann Michel-Kerjan, “What Drives Households to Buy Flood Insurance? New Evidence From Georgia,” *Ecological Economics*, vol. 117 (September 2015), pp. 153–161, <https://doi.org/10.1016/j.ecolecon.2015.06.024>.

56. Caroline Ratcliffe and others, “From Bad to Worse: Natural Disasters and Financial Health,” *Journal of Housing Research*, vol. 29, sup. 1 (November 2020), pp. S25–S53, <https://doi.org/10.1080/10527001.2020.1838172>; and Brigitte Roth Tran and Tamara Lynn Sheldon, “Same Storm, Different Disasters: Consumer Credit Access, Income Inequality, and Natural Disaster Recovery” (SSRN, May 28 2019), <https://dx.doi.org/10.2139/ssrn.3380649>.

57. Joshua Graff Zivin and Matthew Neidell, “Temperature and the Allocation of Time: Implications for Climate Change,” *Journal of Labor Economics*, vol. 32, no. 1 (January 2014), pp. 1–26, <https://doi.org/10.1086/671766>.

58. A. Patrick Behrer and others, “Heat Has Larger Impacts on Labor in Poorer Areas,” *Environmental Research Communications*, vol. 3, no. 9 (September 2021), article 095001, <https://doi.org/10.1088/2515-7620/abffa3>.

damaging. That region has the lowest per capita income in the United States today, which will make it especially vulnerable to financial losses due to climate change.

The economic repercussions for the Southeast will be multifaceted. According to one estimate, if global warming reached 4°C, the Southeast would bear one-third of the national losses in labor productivity due to extreme heat by 2090.⁵⁹ Those losses in productivity will be concentrated in industries that operate outdoors, where adaptations are difficult to implement. The Southeast already experiences higher rates of occupational heat-related deaths in agriculture, construction, and other outdoor sectors.⁶⁰ Decreases in agricultural yields are also expected throughout the South, and the region will see the greatest increases in energy demand in order to adapt to a hotter climate.⁶¹

Finally, the Southeast will be disproportionately affected by the risk of increased flooding and hurricanes.

In Isle de Jean-Charles, Louisiana, for instance, former residents have already been forced to abandon their land because of sea-level rise. Other residents of Gulf Coast counties have seen large increases in rates for homeowner's insurance as insurers react to growing climate risks. And as land in the western Gulf Coast continues to sink because of river sediment compaction and groundwater, oil, and mineral extraction, that region will face the nation's fastest rate of sea-level rise, with sea-level increases 1.0 foot to 1.5 feet greater than the U.S. average by 2100.⁶²

Other regions will also experience significant effects of climate change specific to their area. Those effects will include the following:

- In the Southwest, rising temperatures will be accompanied by drier weather and declining snowpacks, which will reduce available water for cities, farms, and hydroelectric power plants.
- Along the Mississippi River, increases in precipitation in the spring will raise the risk of floods, whereas higher temperatures in the summer will reduce water levels and threaten transportation.
- In Alaska, thawing permafrost will sink or collapse, posing threats to any infrastructure and buildings on top of it.

59. Environmental Protection Agency, *Multi-Model Framework for Quantitative Sectoral Impacts Analysis: A Technical Report for the Fourth National Climate Assessment*, EPA 430-R-17-001 (May 2017), <https://tinyurl.com/m4jjd4k3>.

60. Diane M. Gubernot, G. Brooke Anderson, and Katherine L. Hunting, "Characterizing Occupational Heat-Related Mortality in the United States, 2000–2010: An Analysis Using the Census of Fatal Occupational Injuries Database," *American Journal of Industrial Medicine*, vol. 58, no. 2 (February 2015), pp. 203–211, <https://doi.org/10.1002/ajim.22381>.

61. Solomon Hsiang and others, "Estimating Economic Damage From Climate Change in the United States," *Science*, vol. 356, no. 6345 (June 30, 2017), pp. 1362–1369, <https://doi.org/10.1126/science.aal4369>.

62. William V. Sweet and others, *Global and Regional Sea Level Rise Scenarios for the United States: Updated Mean Projections and Extreme Water Level Probabilities Along U.S. Coastlines*, Technical Report NOS 01 (National Oceanic and Atmospheric Administration, February 2022), <https://tinyurl.com/cmsv48xr>.

Appendix: Research About Climate Risks to the United States

To develop its analysis of the risks of climate change to the United States, the Congressional Budget Office drew on several intersecting bodies of research. This appendix presents CBO's sources, organized by topic and in reverse chronological order. Studies that address multiple topics are listed under each one.

Climate Change

Weather Patterns and Sea-Level Rise

William V. Sweet and others, *Global and Regional Sea Level Rise Scenarios for the United States: Updated Mean Projections and Extreme Water Level Probabilities Along U.S. Coastlines*, Technical Report NOS 01 (National Oceanic and Atmospheric Administration, February 2022), <https://tinyurl.com/cmsv48xr>.

Richard P. Allan and others, "Summary for Policymakers," in Valérie Masson-Delmotte and others, eds., *Climate Change 2021: The Physical Science Basis, Working Group I Contribution to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* (Cambridge University Press, 2021), pp. 3–32, <https://doi.org/10.1017/9781009157896>.

Deliang Chen and others, "Framing, Context, and Methods," in Valérie Masson-Delmotte and others, eds., *Climate Change 2021: The Physical Science Basis, Working Group I Contribution to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* (Cambridge University Press, 2021), pp. 147–286, <https://doi.org/10.1017/9781009157896>.

June-Yi Lee and others, "Future Global Climate: Scenario-Based Projections and Near-Term Information," in Valérie Masson-Delmotte and others, eds., *Climate Change 2021: The Physical Science Basis, Working Group I Contribution to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* (Cambridge University Press, 2021), pp. 553–672, <https://doi.org/10.1017/9781009157896>.

Josep G. Canadell and others, "Global Carbon and Other Biogeochemical Cycles and Feedbacks," in Valérie Masson-Delmotte and others, eds., *Climate Change 2021: The Physical Science Basis, Working Group I Contribution to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* (Cambridge University Press, 2021), pp. 673–816, <https://doi.org/10.1017/9781009157896>.

Baylor Fox-Kemper and others, "Ocean, Cryosphere, and Sea Level Change," in Valérie Masson-Delmotte and others, eds., *Climate Change 2021: The Physical Science Basis, Working Group I Contribution to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* (Cambridge University Press, 2021), pp. 1211–1362, <https://doi.org/10.1017/9781009157896>.

Flavio Lehner and others, "Partitioning Climate Projection Uncertainty With Multiple Large Ensembles and CMIP5/6," *Earth System Dynamics*, vol. 11, no. 2 (May 2020), pp. 491–508, <https://doi.org/10.5194/esd-11-491-2020>.

Natural Disaster Risks

Congressional Budget Office, *Flood Damage and Federally Backed Mortgages in a Changing Climate* (November 2023), www.cbo.gov/publication/59379.

Oliver E. J. Wing and others, "Inequitable Patterns of U.S. Flood Risk in the Anthropocene," *Nature Climate Change*, vol. 12, no. 2 (February 2022), pp. 156–162, www.nature.com/articles/s41558-021-01265-6.

S. A. Parks and J. T. Abatzoglou, "Warmer and Drier Fire Seasons Contribute to Increases in Area Burned at High Severity in Western U.S. Forests From 1985 to 2017," *Geophysical Research Letters*, vol. 47, no. 22 (November 28, 2020), <https://doi.org/10.1029/2020GL089858>.

John T. Abatzoglou and A. Park Williams, “Impact of Anthropogenic Climate Change on Wildfire Across Western U.S. Forests,” *Proceedings of the National Academy of Sciences of the United States of America*, vol. 113, no. 42 (October 10, 2016), pp. 11770–11775, <http://doi.org/10.1073/pnas.1607171113>.

David M. Roms and others, “Projected Increase in Lightning Strikes in the United States due to Global Warming,” *Science*, vol. 346, no. 6211 (November 14, 2014), pp. 851–854, <http://doi.org/10.1126/science.1259100>.

Tipping Points

Robert E. Kopp and others, “‘Tipping Points’ Confuse and Can Distract From Urgent Climate Action,” *Nature Climate Change* (December 3, 2024), <https://doi.org/10.1038/s41558-024-02196-8>.

Nico Wunderling and others, “Climate Tipping Point Interactions and Cascades: A Review,” *Earth System Dynamics*, vol. 15, no. 1 (January 2024), pp. 41–74, <https://doi.org/10.5194/esd-15-41-2024>.

Jesse F. Abrams and others, “Committed Global Warming Risks Triggering Multiple Climate Tipping Points,” *Earth’s Future*, vol. 11, no. 11 (November 2023), article e2022EF003250, <https://doi.org/10.1029/2022EF003250>.

Seaver Wang and others, “Mechanisms and Impacts of Earth System Tipping Elements,” *Reviews of Geophysics*, vol. 61, no. 1 (March 2023), article e2021RG000757, <https://doi.org/10.1029/2021RG000757>.

Timothy M. Lenton and others, eds., *Global Tipping Points: Report 2023* (University of Exeter, 2023), <https://report-2023.global-tipping-points.org>.

David I. Armstrong McKay and others, “Exceeding 1.5°C Global Warming Could Trigger Multiple Climate Tipping Points,” *Science*, vol. 377, no. 6611 (September 9, 2022), article eabn7950, <https://doi.org/10.1126/science.abn7950>.

Organisation for Economic Co-operation and Development, *Climate Tipping Points: Insights for Effective Policy Action* (2022), <https://tinyurl.com/bdz6wrfe>.

Deliang Chen and others, “Framing, Context, and Methods,” in Valérie Masson-Delmotte and others, eds., *Climate Change 2021: The Physical Science Basis, Working Group I Contribution to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* (Cambridge University Press, 2021), pp. 147–286, <https://doi.org/10.1017/9781009157896>.

June-Yi Lee and others, “Future Global Climate: Scenario-Based Projections and Near-Term Information,” in Valérie Masson-Delmotte and others, eds., *Climate Change 2021: The Physical Science Basis, Working Group I Contribution to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* (Cambridge University Press, 2021), pp. 553–672, <https://doi.org/10.1017/9781009157896>.

Josep G. Canadell and others, “Global Carbon and Other Biogeochemical Cycles and Feedbacks,” in Valérie Masson-Delmotte and others, eds., *Climate Change 2021: The Physical Science Basis, Working Group I Contribution to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* (Cambridge University Press, 2021), pp. 673–816, <https://doi.org/10.1017/9781009157896>.

Baylor Fox-Kemper and others, “Ocean, Cryosphere, and Sea Level Change,” in Valérie Masson-Delmotte and others, eds., *Climate Change 2021: The Physical Science Basis, Working Group I Contribution to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* (Cambridge University Press, 2021), pp. 1211–1362, <https://doi.org/10.1017/9781009157896>.

Effects on Economic Output

Ishan B. Nath, Valerie A. Ramey, and Peter J. Klenow, *How Much Will Global Warming Cool Global Growth?* Working Paper 32761 (National Bureau of Economic Research, July 2024), www.nber.org/papers/w32761.

Council of Economic Advisers and Office of Management and Budget, *Assessing Methods to Integrate the Physical Risks and Transition Risks and Opportunities of Climate Change Into the President’s Macroeconomic Forecast* (April 2024), <https://tinyurl.com/wbkeprh8>.

Council of Economic Advisers and Office of Management and Budget, *Methodologies and Considerations for Integrating the Physical and*

Transition Risks of Climate Change Into Macroeconomic Forecasting for the President's Budget (March 2023), <https://tinyurl.com/3zek6v33>.

Council of Economic Advisers and Office of Management and Budget, *Climate-Related Macroeconomic Risks and Opportunities* (April 2022), <https://tinyurl.com/4v3nnns9>.

Richard G. Newell, Brian C. Prest, and Steven E. Sexton, "The GDP–Temperature Relationship: Implications for Climate Change Damages," *Journal of Environmental Economics and Management*, vol. 108 (July 2021), article 102445, <https://doi.org/10.1016/j.jeem.2021.102445>.

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List of Figures

1-1.	Projected Global Carbon Dioxide Emissions Through 2100	7
1-2.	Projected Concentrations of Carbon Dioxide in the Atmosphere Through 2100	8
1-3.	Projected Global Warming Relative to the 1850–1900 Period Through 2100	9
1-4.	Projected Increases in Temperatures in the Contiguous United States by 2100, by Amount of Global Warming	10
1-5.	Projected Changes in Precipitation in the Contiguous United States by 2100, by Amount of Global Warming	11
1-6.	Projected Increases in Regional Sea Levels by 2100, by Average U.S. Sea-Level Increase	12
1-7.	Changes in Natural Processes That Would Have Major Effects on the Climate	14
1-8.	Ranges of Temperature Thresholds and Time Frames for Major Changes in Climate Processes	15
2-1.	The Distribution of Possible Changes to Real GDP in 2100 From Climate Change	20

About This Document

This report was prepared at the request of the Chairman of the Senate Budget Committee. In keeping with the Congressional Budget Office's mandate to provide objective, impartial analysis, the report makes no recommendations.

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CBO seeks feedback to make its work as useful as possible. Please send any comments to communications@cbo.gov.



Phillip L. Swagel
Director
December 2024

Updates

The Congressional Budget Office has updated this report since its original publication. Following the publication of this report, the journal *Nature* retracted a study that it had published in April 2024. That study, “The Economic Commitment of Climate Change,” had informed some of CBO’s estimates of the effects of climate change on GDP. The estimates in this report have been updated to exclude the results from that study. Both the PDF and online versions of the report were updated, but for ease of reference, this list indicates the locations of the changes in the PDF.

The following changes were made on December 17, 2025:

At a Glance, bullet beginning “Effect on GDP”: CBO’s estimates of effects with a 5 percent chance were changed from a 21 percent decrease in GDP to a 17 percent decrease and from a 6 percent increase in GDP to a 7 percent increase. CBO’s central estimate was changed from a 4 percent decrease in GDP to a 3 percent decrease.

Page 1, right-hand column, second paragraph under “How Will Climate Change Affect GDP?”: CBO’s estimates of effects with a 5 percent chance were changed from a 21 percent decrease in GDP to a 17 percent decrease and from a 6 percent increase in GDP to a 7 percent increase. Expressed in terms of 2023 dollars, CBO’s estimates of those effects were changed from a \$5.7 trillion loss to a \$4.7 trillion loss and from a \$1.6 trillion gain to a \$1.9 trillion gain, respectively. CBO’s central estimate was changed from a 4 percent decrease in GDP to a 3 percent decrease.

Page 20, Figure 2-1: The same changes to estimates reported in the At a Glance were made to the values in the figure and the caption.

Page 20, right-hand column, second paragraph under “Projected Effects on GDP”: The same changes to estimates reported on page 1 were made in this paragraph.

Page 21, right-hand column, first sentence under “CBO’s Analytical Approach”: The number of studies underlying CBO’s projections of the effect of temperature change on GDP was changed from 15 to 14.

Page 45, left-hand column, “Estimates of the Effect of Temperature Change on GDP” section: The reference to the study by Maximilian Kotz, Anders Levermann, and Leonie Wenz (2024) was deleted.